Part IX

Risk, Capital, and Regulation

The last part turns from pricing a trade to holding a book of them. Value at risk and expected shortfall measure the exposure, the capital rules set what it costs, and the mishaps chapter shows where losses actually came from: mostly from what measures and rules cannot see (cash, size, information, the venue) rather than from the instruments. Machine learning, which changes what is computable rather than what it costs, is treated in the online technical note [TN41]. If only one chapter from this part can be assigned, it is Chapter 40.

38 **Value at Risk and Expected Shortfall**
the measures, their backtests, and where each one misleads

39 **Market Risk Capital and Model Risk**
FRTB, the P&L attribution test, and non-modellable factors

40 **Derivatives Mishaps and What We Can Learn**
the cases, and the control that was missing in each

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Chapter 38
Value at Risk and Expected Shortfall

Earlier chapters took one instrument or one market at a time: how it works, who uses it, and what it does to the risks of the people who hold it. This one asks what a whole portfolio of positions can lose. That is a different question, and its honest answer is a distribution of losses rather than one figure.

The difficulty is not computational. A trading book's loss distribution can be simulated to any precision the machine allows. The difficulty is that a risk report, a capital rule, a limit framework, and a margin call each need a single number. Everything in this chapter is a way of choosing which number, and the choices are not equivalent—one of them, the one the industry used for twenty years, can penalize diversification.

Two things make this chapter necessary rather than optional. Chapter 39 sets capital from these measures, so an error here is an error in what a bank must hold. Expected shortfall is also a standard objective in deep hedging, the learned hedging strategies of the online note [TN41], so a reader who does not have it defined cannot read what that work optimizes.

Sections 38.1 to 38.4 define the two measures and the property that separates them. Sections 38.5 to 38.9 compute them by historical simulation, by a model, for books with options, under fat tails, and for losses not yet observed; Sections 38.10 to 38.14 cover the inputs: volatility, correlation, copulas, factor reduction, and scaling across horizons. Sections 38.15 to 38.18 put the measures to use: attribution to positions, backtesting, stress testing, and stressed calibration. Section 38.19 says what the measures cannot see, Section 38.20 who uses them and why, and Section 38.21 where the ideas came from.

38.1 WHAT A RISK MEASURE IS FOR

A **risk measure** maps a loss distribution to a single number. Three uses pull in different directions and it is worth separating them before choosing.

A limit is a control. It must be stable enough that a trader can plan against it and responsive enough to bind when it should, and, compared across desks, it must add up.

Capital is a buffer. It answers "how much loss must be survivable," so it cares about the size of bad outcomes, not their frequency.

A margin requirement is a forecast over a specific horizon, the margin period of risk of Section 8.4, which is set by how long a close-out takes rather than by anyone's risk appetite.

These are three different questions and the market answers them with variants of one idea, a number read off the tail of the loss distribution. A measure that appears to behave strangely is often being asked a question it was not built for.

38.2 VALUE AT RISK

Value at risk (VaR) at confidence c over horizon h is the loss that is exceeded with probability $1 - c$:

$$\boxed{\mathbb{P}\big(L > \mathrm{VaR}_c\big) = 1 - c}, \tag{38.1}$$

where $L = V_0 - V_h = -\Delta V$ is the portfolio's loss over the horizon h, its value now less its value at the horizon's end, positive when value is lost.

It is a quantile, and the whole of its behavior follows from that. It says where the tail begins and nothing whatever about what is in it.

Where the loss distribution jumps, as a bond portfolio's does at default, or is a finite sample, (38.1) can have no solution or many. VaR is then the smallest loss l with $\mathbb{P}(L \le l) \ge c$, the lower quantile, which is how Examples 38.2 and 38.3 read it.

Its appeal was never statistical. A single currency amount, attached to a stated probability, can be put in front of a board. Section 38.20 argues that this is the reason it displaced everything that preceded it, not any property in (38.1).

38.3 EXPECTED SHORTFALL

Expected shortfall (ES) at confidence c is the average loss over the worst $1 - c$ of outcomes. Each level p beyond c has its own VaR, and expected shortfall is their average:

$$\boxed{\mathrm{ES}_c = \frac{1}{1-c}\int_c^1 \mathrm{VaR}_p \, dp}. \tag{38.2}$$

Because it averages *all* the VaRs beyond c, it reads the whole tail rather than one point on it.

When the loss distribution is continuous, (38.2) is the mean loss given that the VaR is exceeded, $\mathbb{E}[L \mid L > \mathrm{VaR}_c]$, a quantity called the conditional tail expectation. Where the distribution jumps at the VaR, as in Example 38.2, the two differ, and (38.2) is the definition, because it is the form that keeps the measure coherent. The same quantity goes by other names: conditional VaR (CVaR) in the optimization literature and in deep hedging, and tail VaR in insurance. [TN38] derives the representation that makes it optimizable. Example 38.1 computes the two measures side by side on a normal book.

Example 38.1 The two measures on the same book

Consider a \$100 million position with a 2.0% daily volatility (a daily standard deviation of \$2.0 million) and normally distributed returns. Under normality both measures are multiples of the standard deviation σ: $\mathrm{VaR}_c = z\sigma$ and $\mathrm{ES}_c = \sigma N'(z)/(1-c)$, where $z = N^{-1}(c)$ is the standard normal quantile at confidence c and N' is the standard normal density:

Confidence	z	VaR	ES	ES/VaR
95.0%	1.6449	\$3.29m	\$4.13m	1.254
97.5%	1.9600	\$3.92m	\$4.68m	1.193
99.0%	2.3263	\$4.65m	\$5.33m	1.146
99.9%	3.0902	\$6.18m	\$6.73m	1.090

The ratio falls as the confidence rises, because the normal tail thins so fast that conditioning on being past a distant point adds little. That ratio is a property of the normal distribution and not of the portfolio, which is exactly why it cannot be trusted: on a fatter tail it widens, and Example 38.6 shows by how much.

The regulatory switch reads directly off this table. The Basel framework of Chapter 39 replaced 99% VaR with 97.5% ES, and under normality the two nearly coincide:

$$\mathrm{VaR}_{99\%} = 2.3263\,\sigma, \qquad \mathrm{ES}_{97.5\%} = 2.3378\,\sigma, \qquad \text{a ratio of } 1.0049.$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

The confidence level was chosen to make the change roughly capital-neutral for a normal book, so that what moved was not the level of the requirement but its sensitivity to the shape of the tail. Taken on its own, the switch from VaR to ES charges a bank with a genuinely fat-tailed book more and a bank with a normal book the same; the framework changed other things at the same time, which Chapter 39 sets out. Nor does it buy precision. Estimated from 500 days, a 97.5% ES averages only about a dozen widely dispersed losses, and on a fat-tailed book its standard error is about 14% of its value ([TN38]), no better than that of the 99% VaR it replaced, which Section 38.5 computes.

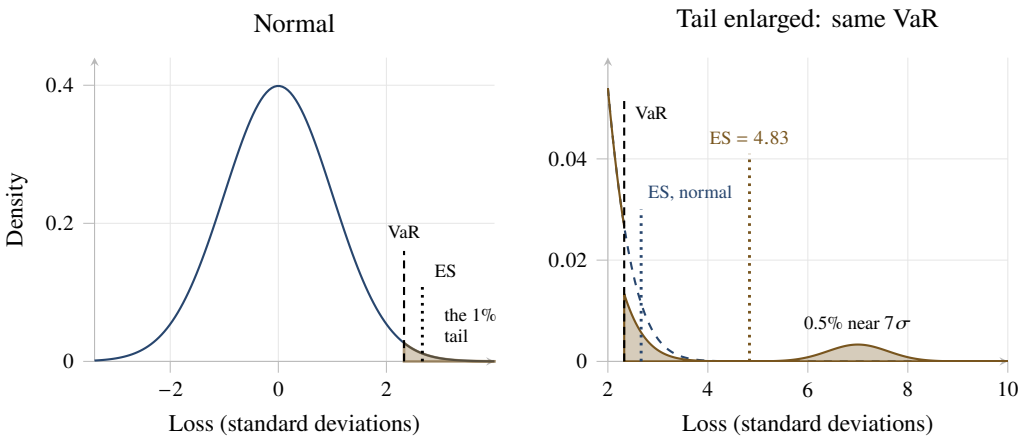


Figure 38.1: Losses at 99%, in standard deviations. Left: the standard normal. Right, the tail enlarged: the same distribution below VaR, with half its 1% tail moved to a cluster near 7σ. Illustrative.

Figure 38.1 carries the whole distinction. VaR is the dashed line: it locates the shaded region's left edge and is indifferent to everything inside it. Two portfolios with identical VaR can have any expected shortfall above that VaR, because the shading can be redistributed arbitrarily without moving the edge. The right panel does exactly that: the two distributions agree up to the VaR, so the VaR is the same 2.3263σ, but half of the 1% tail has been moved to a cluster of losses near 7σ (the profile of a book that has sold deep out-of-the-money protection), and the expected shortfall nearly doubles, from 2.67σ to 4.83σ, the average of the half tail left in place and the cluster at 7σ. A trader whose limit is set in VaR can exploit that, and Section 38.4 shows the exploitation is not even subtle.

38.4 COHERENCE, AND WHERE VAR FAILS

A risk measure ρ is called **coherent** if it has four properties, each a piece of common sense about risk:

- **Monotonicity.** A position that loses more than another in every outcome carries at least as much risk.
- **Positive homogeneity.** Doubling a position doubles its risk.
- **Translation invariance.** Adding an amount of cash to a position reduces its risk by exactly that amount.
- **Subadditivity.** Two positions A and B held together carry no more risk than the two carry separately:

$$\rho(A + B) \;\le\; \rho(A) + \rho(B). \tag{38.3}$$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Subadditivity, the property that matters here, is the formal statement that diversification cannot hurt. It is also what makes a measure usable as a limit: if (38.3) fails, desk limits do not aggregate, and a firm can be within every limit and over its own total.

Expected shortfall satisfies (38.3) always; [TN38] proves this and the other three properties. VaR does not, and the counterexample needs no exotic structure at all: Example 38.2 builds it from two bonds.

Example 38.2 Two bonds that break the measure

A book holds two independent corporate bonds, \$10 million each, with a 0.8% default probability and zero recovery. Measure at 99%.

Each bond alone. A loss occurs with probability 0.008, which is *less* than the 1% tail being measured. The no-default branch carries 99.2% of the probability, so the quantile is zero up to that level and jumps to the full loss beyond it:

$$\mathrm{VaR}_c(\text{one bond}) = \begin{cases} \$0, & c \le 99.2\%, \\ \$10\text{m}, & c > 99.2\%. \end{cases}$$

At 99% the confidence level sits just below the jump, so $\mathrm{VaR}_{99\%}(\text{one bond}) = \$0$.

The two together. The loss distribution is:

Loss	Probability	Cumulative
\$0m	0.984064	0.984064
\$10m	0.015872	0.999936
\$20m	0.000064	1.000000

Since $0.984064 < 0.99$, the 99th percentile falls in the \$10m outcome:

$$\mathrm{VaR}_{99\%}(\text{both}) = \$10\text{m} \;>\; \$0 + \$0 = \mathrm{VaR}_{99\%}(A) + \mathrm{VaR}_{99\%}(B).$$

Diversifying made the measured risk go from nothing to \$10 million. Nothing pathological was constructed: two ordinary bonds, independent, with a default probability smaller than the tail being measured.

Expected shortfall behaves. It averages the worst 1% rather than reading its boundary. For one bond the worst 1% is the 0.8% default branch at \$10m and 0.2% of the no-loss branch, so $\mathrm{ES}_{99\%} = (0.008 \times 10 + 0.002 \times 0)/0.01 = \$8\text{m}$, against a VaR of zero. For the pair:

$$\mathrm{ES}_{99\%}(\text{both}) = \frac{(1 - 0.999936) \times 20 + (0.999936 - 0.99) \times 10}{0.01} = \$10.06\text{m} \;\le\; \$16\text{m}.$$

The mechanism is worth naming because it generalizes. VaR fails most readily where the loss distribution has a *jump* near the chosen quantile, which describes a credit portfolio of few issuers, every book with digital or barrier payoffs, and every position with a default or knock-out. It can fail on continuous distributions too, if they are skewed or fat-tailed enough, and it is guaranteed to be subadditive at any confidence level above one half when losses are **elliptical**: jointly distributed so that every portfolio's loss has the same shape, differing only in location and scale, as jointly normal and jointly Student's t losses do. So it is well behaved on smooth, roughly elliptical distributions and unreliable on precisely the books that need measuring.

VaR's blindness beyond the quantile also creates a perverse incentive. A desk measured on VaR can sell far-out-of-the-money options, collect premium, and report little or no additional risk, because the large loss has a probability below $1 - c$ and so lies beyond the quantile. Expected shortfall counts that loss at its probability-weighted size, which is a practical argument for the switch, and Section 38.8 gives a sharper one.

38.5 HISTORICAL SIMULATION

The dominant method is also the simplest. Take the last n trading days (500, about two years, is common, and the Basel rules require at least one year) and apply each day's market moves

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

to *today's* portfolio, producing n hypothetical profits and losses. The measure is read off that sample.

What it assumes is one thing and it is a large one: that the last n days are a fair sample of tomorrow. It makes no distributional assumption at all, which is its great advantage: fat tails, skew, and the actual joint behavior of markets are captured without being modeled.

What it cannot do is produce a loss that has not happened. A 500-day window at 99% is estimating from five observations, and it cannot represent a move larger than the largest in the sample. A market that has been calm for two years reports a small number regardless of what is in the book, which is Section 38.18's problem and the reason the Basel capital framework does not rely on this window alone.

Weighting the observations helps but does not solve the problem. Exponentially weighting recent days makes the measure responsive, at the cost of shrinking the effective sample, which a tail estimate can least afford. Example 38.3 reads both measures off a 500-day window and then weights the same window twice, first with the bad days old, then with them recent.

Example 38.3 Five losses out of five hundred

The book is $10 million in an equity index, scored against the last 500 trading days. The 500 daily returns are simulated for illustration, not market data: Student's t with four degrees of freedom, scaled to a 1.25% daily standard deviation, the fat-tailed shape Section 38.8 examines. Each day's return applied to today's $10 million gives one scenario loss. The six worst losses are:

		As recorded		Calendar reversed	
Rank	Loss ($000)	Days ago	Weight	Days ago	Weight
1	581	428	0.064%	73	0.379%
2	437	458	0.055%	43	0.441%
3	424	446	0.059%	55	0.415%
4	408	249	0.157%	252	0.155%
5	396	442	0.060%	59	0.407%
6	357	383	0.080%	118	0.303%

With equal weights each scenario carries 0.2%, so the five worst losses are exactly the 1% tail. The 99% VaR is the loss with exactly five scenarios beyond it, the sixth worst, which is how (38.1) reads on a sample of 500: $\text{VaR}_{99\%} = \$357{,}000$. Expected shortfall is the mean of the five losses beyond it:

$$\text{ES}_{99\%} = \frac{581 + 437 + 424 + 408 + 396}{5} = \$449{,}000.$$

Some systems report the fifth-worst loss, $396,000, instead. That the convention alone moves the number by 11% is a measure of how little five observations are.

Now weight by recency, giving the scenario i days ago the weight

$$\pi_i = \frac{\lambda^i}{\sum_{j=1}^{500} \lambda^j}, \qquad \lambda = 0.995,$$

so that yesterday counts 0.544% and the oldest day 0.045%. The VaR is then the first loss, counting down from the worst, at which the running weight passes 1%. Four of the five worst losses are a cluster 428 to 458 days old, and the five together carry only 0.394%. The running weight does not pass 1% until the twelfth-worst loss, and the weighted VaR is $287,000, 20% below the equal-weighted figure, because the window's bad episode is old. Reverse the calendar, so that the same cluster is 43 to 73 days old, and the three worst losses alone carry 1.24%: the weighted VaR is the third-worst loss, $424,000, 19% above.

The weights moved the VaR between $287,000 and $424,000 without adding a scenario. They can only choose among losses already in the window, so whatever the weights, neither measure can

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

exceed \$581,000, the largest loss in the sample. A window whose worst day was mild reports a mild number however its days are weighted.

The weighting in the example is the scheme of Boudoukh, Richardson and Whitelaw (1998). A second refinement rescales the scenarios to today's volatility instead of reweighting them, and Section 38.10 takes it up once the volatility estimators it needs are in place.

Five observations are also a small sample in a precise sense. An estimated quantile has a sampling standard error of approximately

$$\text{s.e.}\big(\widehat{\text{VaR}}_c\big) \approx \frac{1}{f(\text{VaR}_c)}\sqrt{\frac{c\,(1-c)}{n}}.$$

The square root is the standard error of a proportion: the fraction of the n days that land beyond the estimate is uncertain by that much. Dividing by $f(\text{VaR}_c)$, the density of the loss at the quantile, converts that uncertainty in probability into uncertainty in dollars, and the tail is where the density is lowest, so a small error in probability becomes a large error in the loss.

For the book of Example 38.3, the daily loss on the \$10 million position has the example's 1.25% standard deviation, \$125 thousand, and is a standard t_4 variable rescaled to it. A standard t_4 has variance 2, so the scale, in thousands of dollars, is $125 \times \sqrt{1/2} = 88.39$. The t_4's 99% quantile is 3.7469, and its density is $f_4(t) = \frac{3}{8}(1 + t^2/4)^{-5/2}$. The loss's own density is that density divided by 88.39: stretching a variable by a factor spreads the same probability over a range that much wider, so the probability per thousand dollars is that much smaller. With $c = 0.99$ and $n = 500$, the calculation runs

$$\begin{aligned}
\sqrt{c\,(1-c)/n} &= \sqrt{0.99 \times 0.01/500} = 0.004450,\\
\text{VaR}_{99\%} &= 88.39 \times 3.7469 = \$331.2 \text{ thousand},\\
f(\text{VaR}_{99\%}) &= \frac{f_4(3.7469)}{88.39} = \frac{0.008682}{88.39} = 0.0000982 \text{ per thousand dollars},\\
\text{s.e.} &= \frac{0.004450}{0.0000982} = \$45.3 \text{ thousand},
\end{aligned}$$

14% of the true VaR. The example's \$357,000 is a little over half a standard error from the truth, well within what a 500-day window can resolve.

In practice the density is not known. The **bootstrap** estimates the standard error without it: draw many new samples of 500 from the window itself, with replacement, recompute the VaR on each, and take the standard deviation of the results. For a quantile this far out the answer is rough, because every resample is built from the same few tail losses and none can contain a loss larger than the largest of them.

38.6 THE MODEL-BUILDING APPROACH

The alternative assumes a distribution and computes. For a portfolio linear in its risk factors, with position vector $\mathbf{w}$ and factor covariance Σ,

$$\text{VaR}_c = z\sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}, \tag{38.4}$$

where z is the standard normal quantile at confidence c.

Equation (38.4) is fast, transparent, and differentiable, which is why it survives despite its assumptions: Section 38.15 needs that derivative and historical simulation does not supply one cleanly. Example 38.4 applies it to two stocks.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 38.4 Two stocks

A book holds \$10 million of stock 1, with a 2.0% daily volatility, and \$5 million of stock 2, with 1.0%; the correlation between them is 0.3. In dollars the daily standard deviations are \$0.200m and \$0.050m, and the square root in (38.4) is the portfolio's daily standard deviation:

$$\sigma_P = \sqrt{0.200^2 + 0.050^2 + 2 \times 0.3 \times 0.200 \times 0.050} = \sqrt{0.0485} = \$0.2202\text{m}.$$

At 99%, $z = 2.3263$, so the one-day VaR is $2.3263 \times 0.2202 = \$0.512$m. Over ten days the square-root-of-time rule of Section 38.14 multiplies by $\sqrt{10}$, giving \$1.620m. It is a rule, not a result: it holds only if daily returns are independent, and that section shows what happens when they are not.

Standalone, the two positions have VaRs of $2.3263 \times 0.200 = \$0.465$m and $2.3263 \times 0.050 = \$0.116$m, which sum to \$0.582m. The portfolio's \$0.512m is \$0.069m less, a diversification benefit of 11.9% that exists only because the correlation is below one.

Equation (38.4) makes two assumptions, and both are about shape. It assumes normality, which Example 38.6 measures; and it assumes the book is linear in its factors, which fails for any book with options in it. The usual repair for the second is a **delta–gamma** expansion, which adds a quadratic term so that convexity is at least approximated. Example 38.5 applies it.

Example 38.5 Sold puts on the same stock

Keep the book of Example 38.4, but hold the position in stock 1 through sold put options instead, with the same dollar delta of \$10 million and a dollar gamma of −\$100 million. A sold put gains as the stock rises, so its delta is positive, and like any sold option it has negative gamma. (The dollar delta $\Delta_\$$ and dollar gamma $\Gamma_\$$ are the delta times the stock price and the gamma times its square.) A return r on the stock then changes the position's value by approximately

$$\Delta V \approx \Delta_\$ \, r + \frac{1}{2} \Gamma_\$ \, r^2 = 10\, r - 50\, r^2 \qquad (\$ \text{ millions}).$$

This is Chapter 18's expansion, equation (18.1), restricted to its stock terms and written per unit return: with $\Delta S = S\, r$, its delta term $\Delta\, \Delta S$ becomes $\Delta_\$ \, r$ and its gamma term becomes $\frac{1}{2}\Gamma_\$ \, r^2$. Taken alone, the stock's 99% fall is $r = -2.3263 \times 2.0\% = -4.653\%$. The linear term loses \$0.4653m, the standalone figure above; the quadratic term adds $50\, r^2 = \$0.1082$m, for a loss of \$0.5735m, 23% more than the linear term alone. The loss grows as the stock falls, and a rise produces a loss only beyond 20%, ten standard deviations away, so the loss at the 99% fall is the position's 99% VaR under the expansion. Equation (38.4) would still report \$0.465m, because it sees the dollar delta and nothing else.

Two terms suffice here because the sold puts' gamma, large as it is, changes little over a 4.65% move, so the parabola built from today's delta and gamma stays close to the position's true value. The approximation weakens when gamma itself changes quickly (an option near expiration with the stock close to its strike) and fails outright at a barrier. Had the puts carried a knock-out barrier 3% below the spot, the position would change form as the stock crossed it, and an expansion built from today's delta and gamma could not see a change it had not yet reached.

A portfolio of barriers and autocallables from Chapters 34 and 35 is therefore not well approximated by two terms, and for those books full revaluation under Chapter 20's Monte Carlo is the only reliable route. Section 38.7 sets out that route, and a shortcut for the smooth option books in between.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

38.7 NONLINEAR BOOKS: CORNISH–FISHER AND MONTE CARLO †

Between the linear book of Example 38.4 and a book of barriers lies most of what a derivatives desk holds: options whose value is curved but smooth. Two methods serve them, one analytical and one by simulation, and Table 38.1 sets them beside the two already met.

Where the quadratic approximation is good, its loss distribution can be summarized without simulation. With normally distributed factors, a delta–gamma loss L, with mean μ and standard deviation σ, has known higher moments too:

- its **skewness** $\gamma_1 = \mathbb{E}[(L-\mu)^3]/\sigma^3$ measures asymmetry: zero for a symmetric distribution, positive when the long tail lies on the loss side;
- its **excess kurtosis** $\gamma_2 = \mathbb{E}[(L-\mu)^4]/\sigma^4 - 3$ measures how much heavier its tails are than a normal distribution's, for which it is zero.

The **Cornish–Fisher expansion** turns these moments into a quantile by correcting the normal one:

$$z_{\mathrm{CF}} = z + \frac{z^2-1}{6}\,\gamma_1 + \frac{z^3-3z}{24}\,\gamma_2 - \frac{2z^3-5z}{36}\,\gamma_1^2, \qquad \mathrm{VaR} \approx \mu + z_{\mathrm{CF}}\,\sigma,$$

where $z = N^{-1}(c)$ is the standard normal quantile at the confidence level, 2.3263 at 99%. For the sold puts of Example 38.5, write the return as $r = 0.02Z$, Z standard normal; the loss $-10r + 50r^2$ is then $-aZ + bZ^2$ with $a = 10 \times 0.02 = 0.2$ and $b = 50 \times 0.02^2 = 0.02$ (\$m), and its moments follow from those of Z:

$$\mu = b = 0.020, \quad \sigma = \sqrt{a^2 + 2b^2} = 0.2020, \quad \gamma_1 = \frac{6a^2b + 8b^3}{\sigma^3} = 0.590, \quad \gamma_2 = \frac{48\,b^2(a^2+b^2)}{\sigma^4} = 0.466.$$

A normal quantile on μ and σ alone gives \$0.490m; the skewness term lifts it to \$0.578m and the fourth-order terms, kurtosis and squared skewness, bring it to \$0.573m, against the example's exact \$0.5735m ([TN38]). The expansion is reliable for moderate skewness like this and can misbehave for strongly nonlinear books, which is where simulation takes over.

Monte Carlo VaR draws the risk factors from their joint distribution many times, values the portfolio in each draw, and reads the quantile and the tail mean off the simulated losses. In **full revaluation** each draw reprices every position with its own pricing model; in **partial simulation** each draw applies the delta–gamma approximation, which is faster and inherits that approximation's limits.

Either way the result carries sampling error, and the standard-error formula of Section 38.5 measures it, with the number of draws M in place of the number of days. For the sold puts, the loss density at the VaR is the stock return's density there divided by the rate at which the loss grows with the return, which is $|-10 + 100\,r| = 14.653$ at $r = -4.653\%$, so the density is

$$f(\mathrm{VaR}_{99\%}) = \frac{N'(2.3263)}{0.02 \times 14.653} = 0.0909 \text{ per million dollars.}$$

With $M = 10{,}000$ draws the standard error is

$$\text{s.e.} = \frac{\sqrt{0.99 \times 0.01/10{,}000}}{0.0909} = \$0.011\text{m},$$

1.9% of the \$0.5735m VaR, and with 100,000 draws it is 0.6%. Unlike a historical window, a simulation can be lengthened as far as the machine allows, so its error is a cost of computing rather than a shortage of data.

The table sets out the trade-off. Below the first row, each step down handles curvature more faithfully and costs more computing; historical simulation, which already reprices in every scenario, alone takes its distribution from data rather than from an assumption, and Monte

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Method	Distribution	Options	Typical users
Historical simulation	the window's, as observed	repriced in each day's scenario	bank trading desks; clearing houses, volatility-scaled
Variance–covariance	normal	delta only	limit and allocation systems; fund VaR limits
Delta–gamma with Cornish–Fisher	normal factors	quadratic approximation	quick checks on smooth option books
Monte Carlo, full revaluation	any that can be simulated	repriced in each draw	exotic and structured books; insurers' internal models

Table 38.1: Four ways to compute VaR or expected shortfall.

Carlo alone can be made as precise as the budget allows. Section 38.20 says why each user ends up where the last column puts it.

38.8 FAT TAILS, AND A COUNTERINTUITIVE RESULT

The normal distribution is a poor description of daily returns: large moves occur far more often than it allows. The standard alternative is **Student's t distribution** with ν degrees of freedom, a bell curve whose density falls off as a power of the loss, $|x|^{-(\nu+1)}$, rather than exponentially in its square. The fewer the degrees of freedom, the fatter the tails, and as ν grows the t approaches the normal. Its variance is $\nu/(\nu-2)$ for $\nu > 2$, so a t variable multiplied by $\sqrt{(\nu-2)/\nu}$ has unit variance and can then be scaled to any chosen standard deviation; for $\nu = 4$, a common fit to daily equity returns, the factor is $\sqrt{1/2}$, which is where Section 38.5's 88.39 came from. Example 38.6 uses it to change a book's shape while holding its volatility fixed.

Example 38.6 The same volatility, a different shape

Take the \$100 million book of Example 38.1 and change nothing except the distribution: Student's t with four degrees of freedom, rescaled to the *identical* 2.0% daily standard deviation. The mean and volatility are the same; only the tail is fatter.

	VaR		ES		
Conf.	Normal	Fat-tailed	Normal	Fat-tailed	ES rise
95.0%	\$3.29m	\$3.01m	\$4.13m	\$4.53m	+10%
99.0%	\$4.65m	\$5.30m	\$5.33m	\$7.38m	+39%
99.9%	\$6.18m	\$10.14m	\$6.73m	\$13.70m	+103%

Figure 38.2 draws the two densities. The t is taller in the middle and thinner in the shoulders, and on the logarithmic scale of the right panel its tail lies orders of magnitude above the normal one; the vertical lines mark the two 99.9% VaRs, \$6.18m and \$10.14m.

Read the 95% row again. The fat-tailed VaR is *lower* than the normal one: \$3.01m against \$3.29m, some 8% *less* risk by that measure. Fat tails do not raise VaR at every confidence level. A distribution with the same variance and a heavier tail must be thinner somewhere, and at 95% that is where the quantile sits.

So a bank that fits a fat-tailed model and measures at 95% reports a risk reduction. As a quantile the number is right, but it ignores the tail, and the report does not say so.

Expected shortfall largely avoids this. It rises at every level in the table (10%, 39%, 103%) because it integrates the tail rather than sampling one point of it. It too falls below the normal at a low enough level, but its crossover sits near 89% against VaR's near 97.5%, so at the levels where risk is actually measured it moves the right way. This is a sharper argument for expected shortfall

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

than subadditivity, because it needs no jump in the distribution: it happens on the most ordinary alternative distribution anyone would fit. It also shows what the normal assumption costs where it matters: at 99.9%, the level at which credit and default-risk capital is set, the normal model understates the true expected shortfall by a factor of two.

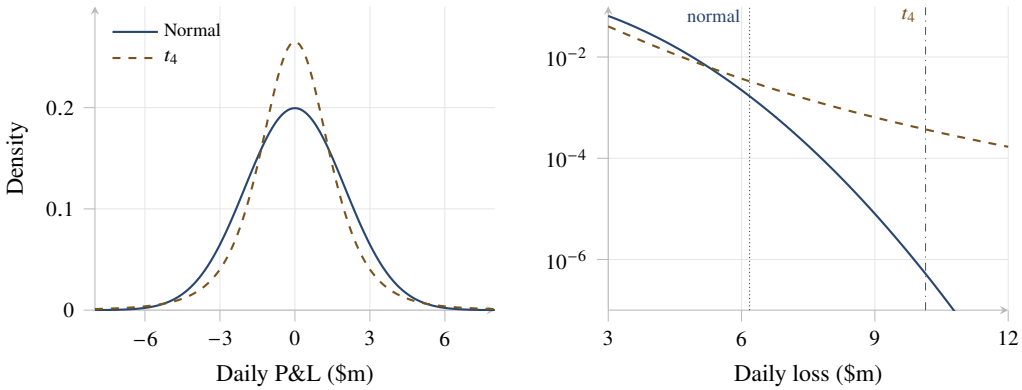


Figure 38.2: Daily P&L of the \$100m book under a normal and a t_4 with the same 2.0% standard deviation. Left: the densities. Right: the loss tail on a log scale, with each model's 99.9% VaR marked.

38.9 ESTIMATING A LOSS THAT HAS NOT HAPPENED †

Section 38.5 left a problem unanswered. Historical simulation cannot report a loss larger than the largest in its window, and a 99.9% measure of a daily loss asks about a percentile where a 500-day sample has half an observation.

Extreme value theory is the standard answer, and Example 38.7 measures what it achieves. The result that the excesses over a high threshold of almost any distribution follow a **generalized Pareto** law lets the tail be *fitted* from the observations that exist and then *extrapolated* past them.

Fix a high threshold u and look only at the losses beyond it. As u rises, the excess $Y = L - u$ of a loss over the threshold has, for almost any underlying distribution, approximately the distribution function

$$\mathbb{P}(Y \le y) = 1 - \left(1 + \frac{\xi\, y}{\psi}\right)^{-1/\xi},$$

with a scale ψ and a shape ξ that carries the tail's thickness:

- $\xi > 0$ is a power-law tail (a Student's t with ν degrees of freedom has $\xi = 1/\nu$, so the book of Example 38.6 has $\xi = 0.25$);
- $\xi = 0$ is an exponential-type tail, the limit for the normal, the lognormal and most other unbounded distributions whose tails die off faster than any power;
- $\xi < 0$ is a tail with an end point.

Fitting ψ and ξ to the losses above u and reading the quantile off the fitted law is the **peaks-over-threshold** method, and Example 38.7 sets it beside the two simpler estimates. [TN38] derives the results it rests on, from the threshold limit to the formulas the example uses.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 38.7 Three ways to estimate a quantile you cannot observe

Take the fat-tailed book of Example 38.6: a daily loss in millions of dollars, distributed as a Student's t with four degrees of freedom and scaled to a standard deviation of 2.0, so that the true 99.9% VaR is \$10.14m and the expected shortfall \$13.70m. A risk manager sees only 500 days of losses drawn from it. One simulated window of 500 has mean loss $\bar{L} = -\$0.0011$m and standard deviation $s = \$1.7953$m, and its 26 largest losses, ranked from the largest down, are:

Ranks 1–5	6–10	11–15	16–20	21–26
7.2509	5.4179	3.7229	3.1511	3.0356
6.8962	4.6913	3.7217	3.1395	3.0329
6.1266	4.4446	3.5340	3.0841	3.0045
5.9353	4.1928	3.4639	3.0812	2.9119
5.7588	4.1022	3.3718	3.0789	2.8497
				2.8481

These losses, with $\bar{L}$ and s, are all the data the three estimates below use.

Normal fit. With $z = N^{-1}(0.999) = 3.0902$ from Example 38.1,

$$\text{VaR} = \bar{L} + z\,s = -0.0011 + 3.0902 \times 1.7953 = \$5.55\text{m}.$$

Empirical quantile. Placing the kth smallest of the 500 losses at percentile $(k-1)/499$ puts the 99.9th percentile at $k = 1 + 0.999 \times 499 = 499.501$, 50.1% of the way from the second-largest loss to the largest:

$$6.8962 + 0.501 \times (7.2509 - 6.8962) = \$7.07\text{m}.$$

Peaks over threshold. Set the threshold at the 26th-largest loss, $u = \$2.8481$m, so that 25 losses, the worst 5%, exceed it. The generalized Pareto law is fitted to their excesses y_i over u by **maximum likelihood**: choose the ψ and ξ under which the observed excesses are most probable, which means maximizing the logarithm of their joint density,

$$\ln \mathcal{L}(\psi, \xi) = -25 \ln \psi - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{25} \ln\left(1 + \frac{\xi\, y_i}{\psi}\right),$$

over $\psi > 0$ and $\xi \geq -1/2$ (below $-1/2$ the maximum-likelihood estimates lose their usual large-sample behavior). That gives $\hat{\psi} = 1.1511$ and $\hat{\xi} = 0.0983$. Since 25 of the 500 days exceed u, a probability of $1 - c = 0.001$ is a fraction $(500/25) \times 0.001 = 0.02$ of the excesses, and solving the fitted law for that fraction gives

$$\text{VaR} = u + \frac{\psi}{\xi}\left[\left(\frac{n}{n_u}(1-c)\right)^{-\xi} - 1\right] = 2.8481 + \frac{1.1511}{0.0983}\left(0.02^{-0.0983} - 1\right) = \$8.34\text{m},$$

with $n = 500$ days, $n_u = 25$ exceedances and $c = 0.999$. Averaging the fitted tail beyond that point gives the expected shortfall:

$$\text{ES} = \frac{\text{VaR} + \psi - \xi u}{1 - \xi} = \frac{8.3389 + 1.1511 - 0.0983 \times 2.8481}{1 - 0.0983} = \$10.21\text{m}.$$

All three VaR estimates fall short of the true \$10.14m. The window's largest loss, \$7.25m, is below it, so the empirical quantile cannot reach it, and the 25 excesses suggest a thinner tail than the true one ($\hat{\xi} = 0.10$ against $\xi = 0.25$, or 0.19, the shape of the generalized Pareto law that best approximates the true t_4 excesses at this threshold, which [TN38] derives), so the extrapolation stops short too. Figure 38.3 draws the three against the data.

The figure shows what each method does where the data run out. The points stop at the largest loss, so the empirical quantile has nowhere further to go. The normal fit's tail falls away from the points almost at once and crosses the 0.001 line at \$5.55m. The generalized Pareto law follows the points and continues past them, reaching \$8.34m, but falls away faster than the

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

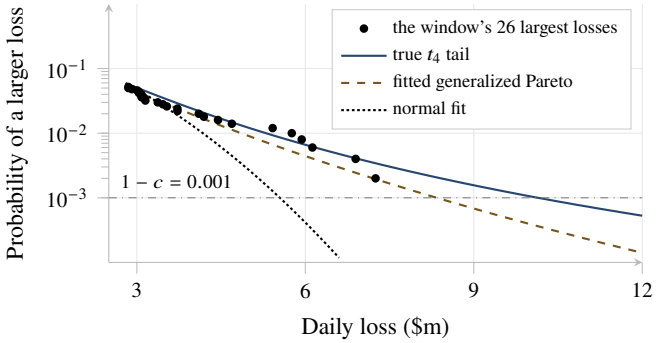


Figure 38.3: The tail of the window in Example 38.7, with the true distribution, the fitted generalized Pareto law and the normal fit. Each curve crosses the 0.001 line at its 99.9% VaR.

true tail, which does not cross until $10.14m.

One window shows how each calculation works, not how well each method performs. The table repeats the three calculations on 400 independent 500-day windows drawn from the same book, of which the example's is the first. The **mean VaR estimate** is the average of a method's 400 answers, the **bias** is that average divided by the true $10.14m, less one, and the **spread** is the standard deviation of the 400 answers.

Method	Mean VaR estimate	Bias	Spread across windows
Empirical quantile	$9.22m	−9.1%	$2.85m
Normal fit	$6.19m	**−39.0%**	**$0.70m**
Peaks over threshold	$10.11m	−0.3%	$3.32m

These are simulation results: the standard error of each mean is its spread divided by $\sqrt{400}$, under 2% of the true VaR for all three methods, and the 400 windows are drawn from a fixed seed. The normal fit is by far the most precise method and it is wrong by 39%. Precision and accuracy point in opposite directions here, and a risk report showing a stable number month after month is displaying the property that a normal assumption is good at producing. Peaks over threshold removes the bias, its −0.3% being well inside its standard error, and pays for it with the widest spread of the three. Its expected shortfall shows the cost: because the formula divides by $1 - \hat{\xi}$, the few windows whose fitted shape comes out high report very large values, so the median window's estimate, $11.48m, falls below the true $13.70m while the mean, $15.19m, is pulled above it.

The ceiling on historical simulation is not hypothetical. A window's largest loss stays below the true 99.9% VaR with probability $0.999^{500} = 61\%$, so in most windows no confidence level gets the empirical quantile to the right answer, because it lies off the end of the sample.

Two cautions keep this from being a solution. The threshold is a choice, and the estimate moves with it: a parameter the analyst picks, in a calculation the analyst is being judged by. And the method extrapolates the tail's *shape* from data in the tail's shoulder, which assumes the shape continues. It cannot see a mechanism that switches on only in the extreme, which is exactly what Section 38.11's correlations and Section 38.12's tail dependence describe.

So extreme value theory converts an impossible estimate into a difficult one. That is a real gain (a bias within sampling error of zero, against 39%), but it is not a known answer.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

38.10 ESTIMATING VOLATILITY

The model-building methods above need a volatility, and historical simulation can use one to rescale its scenarios. Two estimators dominate and they differ in one structural respect.

Exponentially weighted moving average (EWMA) weights past squared returns geometrically:

$$\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1-\lambda)\, r_{n-1}^2, \tag{38.5}$$

where σ_n^2 is the variance estimate for day n, made at the end of day $n-1$, and r_{n-1} is the return on day $n-1$. The usual $\lambda = 0.94$ is the value J.P. Morgan's RiskMetrics (1996) chose because it minimized one-day forecast error across a broad set of daily series. The same recursion updates a covariance, with the product of two assets' returns in place of the squared return, which is how the correlations of Section 38.11 are usually kept current.

The **generalized autoregressive conditional heteroskedastic** (GARCH) model, in its GARCH(1,1) form, adds a constant, and the constant changes everything:

$$\sigma_n^2 = \omega + \alpha\, r_{n-1}^2 + \beta\, \sigma_{n-1}^2, \qquad V_L = \frac{\omega}{1-\alpha-\beta}, \tag{38.6}$$

where V_L is the long-run variance to which the forecast reverts. The three parameters are estimated by maximum likelihood on the return history, as the tail parameters were in Example 38.7: the chosen ω, α and β are those under which the observed returns are most probable given the variances the model assigns them. Setting $\omega = 0$ and $\alpha + \beta = 1$ in (38.6) recovers (38.5), so EWMA is GARCH with no long-run level to return to. That single difference decides the entire shape of the forecast, as Example 38.8 shows.

Example 38.8 One bad day, two forecasts

A market with a 1.00% daily volatility takes a −3% day.

Both estimators react almost identically. In percent squared, with $r_{n-1} = -3$ and GARCH's $\omega = 0.000002$ (0.02 in percent squared), $\alpha = 0.06$ and $\beta = 0.93$,

$$\sigma^2_{\text{EWMA}} = 0.94\times 1.00^2 + 0.06\times 3^2 = 1.48, \qquad \sigma^2_{\text{GARCH}} = 0.02 + 0.06\times 3^2 + 0.93\times 1.00^2 = 1.49,$$

for volatilities of 1.2166% and 1.2207%. A single day raises the estimate by about 22%.

They disagree completely about what happens next. GARCH has $\alpha + \beta = 0.99$ and therefore a long-run daily volatility of $\sqrt{0.000002/0.01} = 1.4142\%$, or 22.4% annualized. Writing σ_n^2 for the variance just estimated, the GARCH forecast of the variance k days later is

$$\mathbb{E}\big[\sigma_{n+k}^2\big] = V_L + (\alpha+\beta)^k\,\big(\sigma_n^2 - V_L\big),$$

so each day closes 1% of the remaining gap to the long-run level, and a deviation from it halves in about 69 trading days. Averaging that forecast over the T days to a horizon gives the variance

$$\bar{\sigma}_T^2 = V_L + \frac{1-(\alpha+\beta)^T}{T\,(1-\alpha-\beta)}\,\big(\sigma_n^2 - V_L\big),$$

which annualized is 19.4% for one day, 19.5% for ten days, 20.2% for three months and 21.4% for a year. EWMA's forecast is 19.3% at every horizon: its best guess for volatility in a year is today's estimate.

The two are fit for different jobs. A one-day margin or limit calculation cannot tell them apart, and EWMA's simplicity wins. Anything with a term structure needs the mean reversion: the volatility term structures of Chapter 19 slope the way they do because volatility reverts, and a model that denies reversion produces a flat term structure that the option market visibly contradicts.

Note also which way the current estimate sits. At 19.4% it is still below the model's 22.4% long-run level, and the GARCH forecast is *rising*. A risk system built on EWMA would report no such tendency, and the omission matters most just when volatility is about to rise.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

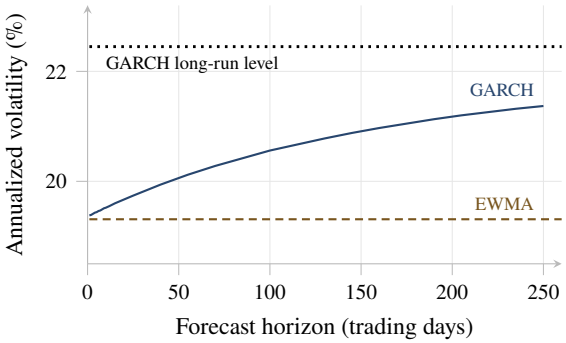


Figure 38.4: Average volatility to the horizon after the shock of Example 38.8.

Figure 38.4 is the argument in one picture. The gap between the two lines is not an estimation disagreement: both estimators saw the same data and agree about today. It is a disagreement about whether volatility has a level, and only one of them can produce a term structure.

Either estimator also repairs historical simulation, which Section 38.5 left reading every past day at the volatility it happened to have. **Volatility-scaled historical simulation**, proposed by Hull and White (1998), multiplies each day's return by $\sigma_{\text{today}}/\sigma_t$, today's volatility estimate over the estimate on the day it occurred. A calm-period day is scaled up when markets are volatile and a crisis day scaled down when they are calm, so the scenarios keep the historical shape at today's scale. Unlike weighting, rescaling can exceed the window's largest loss. **Filtered historical simulation** (Barone-Adesi, Giannopoulos and Vosper 1999) goes a step further: it divides each historical return by that day's GARCH volatility and draws the resulting standardized shocks at random, passing them through the recursion (38.6) to build scenarios over several days. Clearing houses set margin with volatility-scaled scenarios (Section 38.20).

38.11 CORRELATION, AND WHY IT IS THE WEAK INPUT

Example 38.8's estimators supply the diagonal of the covariance matrix in (38.4). The rest of it is harder. A volatility is pinned down by every day's return, large or small; a correlation depends on how the two series move together, and the ordinary days that dominate any window say little about how they move together on a stressed one. Correlations are therefore the least stable input in (38.4). Example 38.9 shows how far the measure moves with one.

Example 38.9 One number, revalued

Position A is \$60m at 1.0% daily volatility; position B is \$40m at 2.5%. In dollars the daily standard deviations are \$0.600m and \$1.000m, and at a correlation of 0.30

$$\sigma_P = \sqrt{0.600^2 + 1.000^2 + 2 \times 0.30 \times 0.600 \times 1.000} = \$1.311\text{m},$$

so the 99% one-day VaR is $2.3263 \times 1.311 = \$3.051$m. Change only the correlation:

Correlation	VaR	Change
−0.50	\$2.028m	−34%
0.00	\$2.713m	−11%
0.30	\$3.051m	—
0.60	\$3.355m	+10%
0.90	\$3.634m	+19%
1.00	\$3.722m	+22%

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

A move from 0.30 to 0.90 raises the reported risk by 19% with no change in any position and no change in any volatility.

Correlations also move most in exactly the episodes the measure exists to describe. Assets that are unrelated in normal conditions become one asset in a liquidation, because the common factor in a crisis is often not economic: it is that the same leveraged holders are selling all of it. So the input is estimated in calm periods and applied in violent ones, and for a book that is long both assets, as this one is, the error has a consistent sign: measured risk is too low.

This is not a call for conservative correlations. It is a reason the framework does not let a bank choose them freely, and Chapter 39's constrained diversification rules exist because of the sensitivity this table shows.

38.12 COPULAS AND TAIL DEPENDENCE

Correlation summarizes dependence in one number, and one number is not enough. Two markets can have identical correlation and behave completely differently in the tail, which is what a copula (Section 24.13) separates out: the marginal distributions on one side, the dependence structure on the other. Example 38.10 holds the margins and the correlation fixed and changes only the copula.

Example 38.10 Same margins, same correlation, different joint extremes

A book holds two exposures, each normally distributed, with correlation 0.50. The only thing that differs is the copula: Gaussian against Student's t with four degrees of freedom.

Probability both exceed	Gaussian copula	t_4 copula	Ratio
95th percentile	0.01219	0.01694	1.39×
99th percentile	0.001294	0.002877	2.22×
99.5th percentile	0.0004963	0.001385	2.79×
99.9th percentile	0.00005426	0.0002635	**4.86**×

The discrepancy grows as the event gets worse, which is the opposite of what a modeler wants from an approximation. The limiting behavior is the cleanest statement. The coefficient of upper tail dependence (the limiting probability that one exposure is extreme given the other is) is exactly zero for a Gaussian copula at every correlation below one, and $2\,t_5\left(-\sqrt{5\times 0.5/1.5}\right) = 0.2532$ for this t copula, with t_5 the distribution function of Student's t with five degrees of freedom. Under a Gaussian copula, the chance that one exposure is in catastrophe given that the other is falls to zero far enough out, however correlated the assets are; under this t copula it never falls below a quarter.

Section 26.6 made this point about CDO tranches, where a Gaussian copula priced the senior tranches as if joint defaults could not cluster in the extreme. Here it decides something more basic: whether a portfolio risk measure gives simultaneous extreme losses their weight. A firm that models dependence with correlations alone has assumed the answer to the question it is asking.

Figure 38.5 shows where the difference lives. It draws 1,500 joint outcomes from each copula, and in the middle the two clouds are hard to tell apart. In the corners they are not: in this draw the t_4 copula puts 26 points beyond both 95th percentiles against the Gaussian's 16, and 7 beyond both 99th percentiles against 1. The two copulas agree in the body and diverge exactly where a risk measure is supposed to be doing its work.

38.13 FACTOR REDUCTION †

A swap book has sensitivities to twenty or more curve points, and (38.4) then needs a 20×20 covariance matrix estimated from data that barely supports it. The standard remedy is to replace the curve points with a few factors: Chapter 32's factor reduction, applied to a risk measure instead of a simulation.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

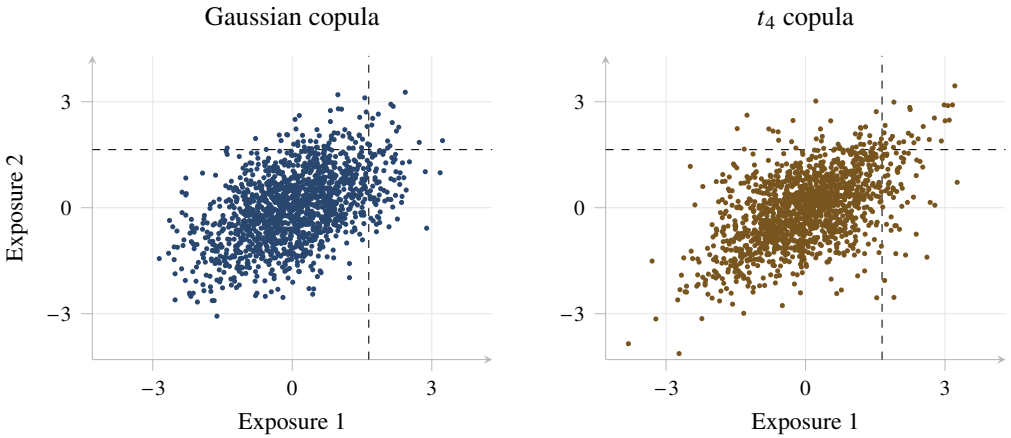


Figure 38.5: Joint outcomes, 1,500 under each copula, in standard deviations, with standard normal margins and correlation 0.50. Dashed lines mark each exposure's 95th percentile.

The factors are the principal components of the curve: the eigenvectors of Σ, the directions of movement that account, in order, for the largest shares of its variance. On a yield curve they are famously interpretable. Example 38.11 shows why interpretable is not the same as sufficient.

Example 38.11 Three factors, two portfolios, two verdicts

The curve has nine tenors from three months to thirty years. Their daily rate-change volatilities, and a directional book's sensitivities to each, are:

Tenor (years)	0.25	0.5	1	2	3	5	7	10	30
Volatility (bp a day)	3.5	4.4	5.3	6.0	6.2	6.0	5.7	5.4	4.8
Directional book ($000 per bp)	10	5	−8	15	12	−10	20	18	6

The correlation between the rate changes at tenors T_i and T_j is

$$\rho_{ij} = 0.7 + 0.3\, e^{-0.3\,|\ln T_i - \ln T_j|},$$

so neighboring tenors move almost together and the three-month and thirty-year rates have a correlation of 0.77. The model is illustrative rather than fitted; its short-end correlations are higher than daily Treasury data would give, which flatters the level factor. The covariance matrix is $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$, and its leading factors are:

Factor	**Variance explained**	**Cumulative**	**Shape**
Level	90.98%	90.98%	all tenors move together
Slope	3.94%	94.92%	short end against long end
Curvature	1.96%	96.88%	wings against the belly
Fourth	1.10%	97.97%	—

Three factors explain 96.88% of the variance of the curve, which is the number usually quoted and the reason three-factor models are standard.

Now measure two books against it. The first is the directional book above, with sensitivities of both signs scattered across the curve. For a vector $\mathbf{w}$ of sensitivities the full-covariance 99% one-day VaR is $2.3263\sqrt{\mathbf{w}^\top \Sigma\, \mathbf{w}}$, as in (38.4), which for this book is $800,607. The second is a **butterfly**, which on a yield curve means a position of one sign at two tenors, the wings, against the opposite sign at a tenor between them, the body. The name is borrowed from the option butterfly

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

of Section 12.3, whose wings and body sit at strikes instead. Sized so that the sensitivities sum to zero, it is flat under a parallel shift and is a position on the curve's curvature, not its level. This one is long \$20,000 per basis point at two years and five years and short \$40,000 at three, with a full VaR of \$117,925. Keeping k factors replaces $\mathbf{w}^\top \Sigma\, \mathbf{w}$ by $\sum_{j \le k} \ell_j (\mathbf{v}_j^\top \mathbf{w})^2$, where ℓ_j and $\mathbf{v}_j$ are the eigenvalues and eigenvectors of Σ, and the table reports each book's VaR together with the share of its own variance that survives. (The VaRs use the unrounded quantile $N^{-1}(0.99) = 2.326348$; with 2.3263 the last digits move by up to \$16.)

	Directional book		Butterfly	
Factors kept	**VaR**	**Variance kept**	**VaR**	**Variance kept**
1	\$795,377	98.7%	\$20,390	3.0%
2	\$797,262	99.2%	\$23,225	3.9%
3	\$797,364	99.2%	\$38,194	10.5%
5	\$797,380	99.2%	\$62,434	28.0%
9	\$800,607	100%	\$117,925	100%

The same three factors that keep 99% of the first book's variance keep 10% of the second's. In VaR terms a three-factor system would report the butterfly at \$38,194 against an actual \$117,925, about a third of its risk, and it would do so while displaying the 96.88% figure as justification.

The two books sit on different parts of the decomposition, and Figure 38.6 shows it. The leading factors are smooth across the curve; the butterfly is three adjacent tenors with alternating signs, a shape none of them resembles, so it projects onto them weakly. The 96.88% is the share of the *curve's* variance in three factors. What a portfolio needs is the share of *its own* variance, and a book constructed to be neutral to level, and nearly neutral to slope as well, has deliberately projected itself onto the directions that were discarded. Factor adequacy is a property of the portfolio, not of the curve—and the portfolios most likely to fail the test are the ones relative-value desks hold.

That gives the practical rule. A risk system may reduce factors, but it must report the residual variance *for each book*, not for the market.

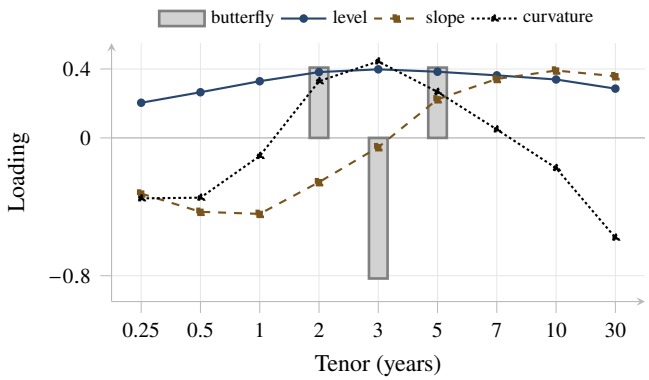


Figure 38.6: Loadings of the three leading factors of Example 38.11 by tenor, and the butterfly's sensitivities scaled to unit length.

38.14 SCALING ACROSS HORIZONS

Every figure so far has been a one-day measure, but capital and limits are often set over longer horizons, and the conversion is usually the square root of time:

$$\mathrm{VaR}_{h\text{-day}} \approx \sqrt{h}\, \mathrm{VaR}_{1\text{-day}}. \tag{38.7}$$

Equation (38.7) requires daily returns that are independent and identically distributed with

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

mean zero, and for the quantile to scale like the standard deviation it also needs them normal; a fat-tailed one-day quantile does not scale by $\sqrt{h}$ even when the days are independent. The first condition is the one this section examines, because no one-day measure, and no one-day backtest, can reveal its failure. Suppose the correlation between returns k days apart is ρ^k, the pattern of a first-order autoregressive process with daily autocorrelation ρ. The variance of an h-day return is then the daily variance times $h + 2\sum_{k=1}^{h-1}(h-k)\rho^k$, one term for each pair of days, so under normality

$$\mathrm{VaR}_{h\text{-day}} = \mathrm{VaR}_{1\text{-day}}\sqrt{h + 2\sum_{k=1}^{h-1}(h-k)\,\rho^k}, \tag{38.8}$$

which reduces to (38.7) when $\rho = 0$. The table applies (38.8) to the one-day 99% VaR of \$4.65m from Example 38.1, scaled to ten days:

Daily autocorrelation	10-day VaR	Error in (38.7)
0.00	\$14.71m	—
0.10	\$16.10m	−9%
0.25	\$18.48m	−20%

The error column is the square-root figure's shortfall as a share of the true ten-day VaR, $(14.71 - \text{true})/\text{true}$: an autocorrelation of 0.10 leaves (38.7) about 9% too low and one of 0.25 a fifth too low, and positive autocorrelation is typical of the forced, one-directional selling of stressed markets, where the number is used. A mean-reverting market, with negative autocorrelation, breaks it the other way, so the rule is not conservative—it is merely wrong in a direction that depends on the market.

The deeper problem is that the horizon should not be a constant at all. Section 8.4's margin period of risk is set by how long a position takes to close, and that is a property of the position: a liquid futures hedge and a bespoke structured trade cannot share a horizon. Chapter 39's liquidity horizons exist to fix precisely this, replacing one number with several by risk-factor class.

38.15 WHERE THE RISK ACTUALLY SITS

A total is not a management tool. The useful question is which position is producing the risk, and the derivative of (38.4) answers it.

Marginal VaR is $\partial\mathrm{VaR}/\partial w_i$, the rate at which the VaR grows as position i grows. Differentiating (38.4),

$$\frac{\partial \mathrm{VaR}}{\partial w_i} = z\,\frac{(\Sigma\mathbf{w})_i}{\sigma_P}, \qquad \sigma_P = \sqrt{\mathbf{w}^\top \Sigma\,\mathbf{w}}, \tag{38.9}$$

where σ_P is the portfolio's standard deviation and $(\Sigma\mathbf{w})_i$ is the covariance of factor i's return with the portfolio's P&L. **Component VaR** is $w_i\,\partial\mathrm{VaR}/\partial w_i$, and because (38.4) is homogeneous of degree one in the positions, the components sum exactly to the total,

$$\sum_i \frac{w_i z(\Sigma\mathbf{w})_i}{\sigma_P} = z\sigma_P,$$

an Euler allocation (Euler's theorem for functions homogeneous of degree one), not an approximation. Example 38.12 applies it to the book of Example 38.9.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Example 38.12 Sixty percent of the money, thirty percent of the risk

	Position	Standalone VaR	Component VaR	Share of risk
A	\$60m (60%)	\$1.396m	\$0.958m	31.4%
B	\$40m (40%)	\$2.326m	\$2.093m	**68.6%**
Total	\$100m	\$3.722m	\$3.051m	100%

B is 40% of the money and 69% of the risk, and the standalone numbers do not show this: they sum to \$3.722m against a portfolio VaR of \$3.051m, an 18.0% diversification benefit that has to be allocated somehow. Component VaR allocates it in the only way that both adds up and agrees with the marginal figures.

The marginal numbers are what a trader should be shown. In return units $\Sigma\mathbf{w} = (0.0090, 0.0295)$ and $\sigma_P = \$1.311\text{m}$, so (38.9) gives $2.3263 \times 0.0090/1.311 = 0.0160$ for A and $2.3263 \times 0.0295/1.311 = 0.0523$ for B. At the margin, a dollar moved into B adds 0.0523 of VaR against A's 0.0160, so shifting \$10m from A to B should add about $10 \times (0.0523 - 0.0160) = \$0.36\text{m}$, or 11.9%; revalued exactly, at \$50m each, the VaR is $2.3263\sqrt{0.500^2 + 1.250^2 + 2 \times 0.30 \times 0.500 \times 1.250} = \$3.441\text{m}$, 12.8% higher, while the book's size is unchanged. A limit on total VaR permits that trade as long as there is room under it; component VaR shows the whole rise came from B.

One caution applies, the chapter's recurring one: an exact allocation says nothing about whether the total is right. Component VaR sums exactly only because VaR is homogeneous; it does not require VaR to be subadditive, and Example 38.2 shows it is not. So the allocation can be exact and the total still wrong. Expected shortfall admits the same Euler decomposition and is coherent; under simulation its components are simply each position's average loss on the scenarios in the tail, $\mathbb{E}[L_i \mid L > \text{VaR}_c]$, which is one reason it is increasingly used for allocation as well.

Incremental VaR answers a third question: how much the VaR changes if a position is removed altogether. In the example, removing B leaves A's standalone \$1.396m, so B's incremental VaR is \$3.051m – \$1.396m = \$1.655m; removing A leaves \$2.326m, so A's is \$0.725m. The two sum to \$2.380m, not to the \$3.051m total, because each credits the whole \$0.671m diversification benefit to the position being removed, so the two together deduct it twice, once more than the total does: $\$3.722\text{m} - 2 \times \$0.671\text{m} = \$2.380\text{m}$. Incremental figures answer whether to keep or close a whole position; component figures are the ones that add up, and marginal figures the ones that guide a small trade.

38.16 BACKTESTING

A risk number is a testable prediction: at 99%, losses should exceed the VaR on 1% of days, and each day they do is an **exception**. Counting exceptions is the most direct validation any of this receives, and Example 38.13 shows how little a year of it establishes.

Example 38.13 Two hundred and fifty days is not much evidence

At 99% over 250 days, the expected number of exceptions is 2.5. With x exceptions in n days, a target exception rate $1 - c = 1\%$ and the observed rate $\hat{p} = x/n$, the Kupiec likelihood-ratio (LR) statistic is

$$\text{LR} = -2 \ln \frac{c^{n-x}\,(1-c)^{x}}{(1-\hat{p})^{n-x}\,\hat{p}^{x}},$$

which under a correct model is approximately chi-squared with one degree of freedom. The test rejects when it exceeds 3.841, the 95th percentile of that distribution, for a nominal 5% significance level; when, as here, few exceptions are expected, the exact binomial probabilities in the table are the alternative:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Exceptions	Probability if correct	Kupiec LR	Reject?	Basel zone
0	0.0811	5.03	**yes**	green
1	0.2047	1.18	no	green
2	0.2574	0.11	no	green
3	0.2149	0.09	no	green
4	0.1341	0.77	no	green
5	0.0666	1.96	no	yellow
7	0.0097	5.50	**yes**	yellow
10	0.0002	12.96	**yes**	red

Anything from one to six exceptions is statistically indistinguishable from a correct model. Those counts are exception rates from 0.4% to 2.4%, so a model whose year looks like 97.6% coverage and one whose year looks like 99.6% both pass as 99% models.

Zero exceptions is a rejection too, and that is the row people forget: a model that never breaches is not conservative but miscalibrated, in the direction that wastes capital. The test is two-sided because a risk measure is a forecast, not a promise. The evidence is thin, though. A correct model shows no exceptions in 8.1% of years, and the 3.841 cutoff, a large-sample approximation that is coarse when only 2.5 exceptions are expected, rejects a correct model 9.5% of the time (zero exceptions, or seven or more) rather than the nominal 5%.

Note where the Basel zones, set in 1996, and the statistics disagree. Five exceptions moves a bank into the yellow zone, where the 1996 rules raised the multiplier applied to its VaR from 3 to between 3.40 and 3.85 (4 in the red zone); the current framework keeps the zones and applies them to a multiplier of 1.5 to 2.0, set out in Chapter 39. Yet at five exceptions the Kupiec statistic is 1.96 and does not reject, and at seven to nine the test rejects while the zone is still yellow. The zones are one-sided: they look only for too many exceptions, and they flag early. A correct model lands in yellow or worse in 10.8% of years, a rate the supervisor accepted in exchange for power: a model whose true coverage is only 97% is flagged in 87% of years, against 63% for a Kupiec rejection.

Figure 38.7 puts the two verdicts on one axis. The shaded band is the range the Kupiec test tolerates and the dashed rules are the Basel boundaries, and they do not line up: a bank with five exceptions has left the green zone and pays a higher multiplier while the statistic sees nothing worth reporting. Note also how much probability mass sits under a correct model at counts a supervisor would treat as a warning.

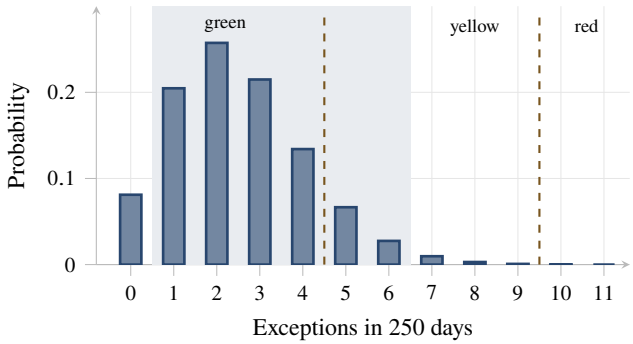


Figure 38.7: Distribution of 99% VaR exceptions in 250 days, Binomial(250, 0.01). Shading marks the range the Kupiec test does not reject; the dashed rules are the Basel zone boundaries.

Counting is only half of the validation. Kupiec asks whether the number of exceptions is right and cannot ask when they arrived, so a model that breaches on five consecutive days in one stressed week passes it exactly as a model that breaches five times spread through the

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

year. Those two models are not equally good, and the second one is the one the bank wants: clustered exceptions mean the model's variance is stale, which is precisely the failure that matters, because it arrives when the losses do.

Christoffersen's conditional-coverage test adds the missing half. Alongside the coverage statistic it tests **independence** (whether an exception yesterday raises the chance of one today) by comparing the observed transition frequencies against the hypothesis that they do not, and the two statistics add, each with one degree of freedom, so the combined statistic is compared with a chi-squared distribution with two degrees of freedom (5.991 at 5%). A model can pass on coverage and fail on independence.

A clean exception count is weaker evidence than it looks. Chapter 39's supervisory framework counts exceptions and does not test their independence, so a desk that wants to know whether its model works has to run that test itself.

Expected shortfall cannot be backtested this way at all. There is no event to count: ES is a conditional mean, so validating it requires estimating a mean from the handful of observations in the tail. Tests built for it exist: Acerbi and Székely (2014) compare each tail loss with the forecast ES on the days VaR was exceeded. A separate obstacle concerns ranking competing forecasts rather than testing one: ES alone is not **elicitable**, meaning that no scoring rule's average is minimized by the true ES, so two ES forecasts cannot be ranked on their record alone; the pair of VaR and ES is elicitable, and [TN38] sets out both. The measure the framework moved to is the one that is harder to check, which is a real cost of the switch and the reason Chapter 39 backtests the VaR while capitalizing the ES.

38.17 STRESS TESTING

Every measure so far answers a question about probability: how large a loss is reached once in a hundred days, and how large the losses are beyond it. A **stress test** answers a different question. It revalues the portfolio under a specified scenario (a set of moves in prices, rates, volatilities, correlations and liquidity) and reports the loss, with no probability attached.

That is its weakness and its use. A number with no likelihood cannot be compared across books the way a VaR can, and it cannot be added to anything. But it does not depend on the window's estimated distribution, so it can describe events the window never contained and joint moves the estimated correlation matrix all but rules out. It targets the gaps Section 38.19 lists.

Scenarios come from two sources. **Historical scenarios** replay the moves of an actual episode on today's positions (October 19, 1987, the autumn of 2008, March 2020, the September 2022 gilt crisis of Section 8.9), and their merit is that nobody can call them implausible. **Hypothetical scenarios** are written by the firm for the risks it actually holds, and a good one moves correlations and liquidity as well as prices, because Section 38.11 showed that the joint behavior is what changes in a crisis. **Sensitivity analysis**, moving one risk factor at a time, is easier to run and to explain, and it misses exactly those joint moves. Example 38.14 sets a hypothetical scenario beside a sensitivity report on the same book.

Example 38.14 Two assets falling together

Take the book of Example 38.9: \$60m in asset A at a 1.0% daily volatility and \$40m in asset B at 2.5%, correlation 0.30, with a 99% one-day VaR of \$3.051m. The scenario is a liquidation day on which both assets fall by four of their own daily standard deviations, A by 4% and B by 10%.

The loss is

$$60 \times 4\% + 40 \times 10\% = \$2.4\text{m} + \$4.0\text{m} = \$6.4\text{m},$$

2.10 times the VaR. A sensitivity report that moves each asset alone by the same four standard deviations shows \$2.4m and \$4.0m, and its worst case, \$4.0m, falls 37.5% short.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Under the model's own assumptions (normal returns, correlation 0.30), the portfolio's daily standard deviation is \$1.311m, so a \$6.4m loss lies 4.88 portfolio standard deviations out. Its probability is about 5 in 10 million a day, once in some 7,500 years of trading days. The stress test does not rely on that figure. It reports the loss and leaves its likelihood to judgment, because the figure is computed from the very correlation the scenario assumes has broken.

Reverse stress testing inverts the question. Instead of asking what a scenario costs, it fixes a loss (the one that would breach a limit, exhaust capital, or end the business) and asks which scenario produces it. It is the exercise most likely to find what the model misses, because it starts from the outcome rather than from the model's risk factors.

For a linear book with normally distributed factors the question has a closed form. Fix the loss at l. Many moves $\mathbf{r}$ of the factors lose exactly l on positions $\mathbf{w}$, and no single one has a probability, but they differ in how plausible they are: the joint density is highest close to the center of the distribution. Distance from the center is measured in standard deviations along whatever direction the move takes, by the **Mahalanobis distance** $\sqrt{\mathbf{r}^\top \Sigma^{-1} \mathbf{r}}$, and the most plausible move is the one that minimizes it subject to $\mathbf{w}^\top \mathbf{r} = -l$. The answer is

$$\mathbf{r}^* = -\, l \, \frac{\Sigma \mathbf{w}}{\mathbf{w}^\top \Sigma \, \mathbf{w}}, \tag{38.10}$$

and it lies l/σ_P standard deviations from the center, where $\sigma_P = \sqrt{\mathbf{w}^\top \Sigma \, \mathbf{w}}$ is the portfolio's standard deviation. Each factor moves in proportion to $(\Sigma \mathbf{w})_i$, its covariance with the portfolio, which is the same vector that gives marginal VaR in (38.9): the scenario loads most heavily on the positions with the highest marginal risk. For a nonlinear book there is no formula, and the scenario has to be found by searching over moves with full revaluation. Example 38.15 applies (38.10) to the book of Example 38.14.

Example 38.15 The most plausible way to lose \$6.4 million

Keep the book of Example 38.14 and fix the loss at the scenario's \$6.4m. In return units, $\Sigma \mathbf{w} = (0.0090,\ 0.0295)$ and $\mathbf{w}^\top \Sigma \, \mathbf{w} = 1.72$, so (38.10) gives

$$r_A^* = -6.4 \times \frac{0.0090}{1.72} = -3.35\%, \qquad r_B^* = -6.4 \times \frac{0.0295}{1.72} = -10.98\%.$$

The loss is $60 \times 3.35\% + 40 \times 10.98\% = \$6.4$m, as required, and the move lies $6.4/1.311 = 4.88$ standard deviations out.

The hand-written scenario of Example 38.14, four standard deviations down on each asset, lies $\sqrt{(4^2 + 4^2 - 2 \times 0.30 \times 4 \times 4)/(1 - 0.30^2)} = 4.96$ standard deviations out by the Mahalanobis distance. The joint density falls as $e^{-d^2/2}$ with that distance d, so the reverse stress test has found a route to the same loss at which the density is $e^{(4.961^2 - 4.880^2)/2} = 1.49$ times as high. It also says where the risk is: on the most plausible path B falls 4.39 of its standard deviations and A only 3.35 of its own, as B's 69% share of the VaR in Example 38.12 suggests. The scenario a risk manager would write by symmetry, both falling equally hard, is not the one that matters most.

Figure 38.8 draws the geometry. The ellipse is the set of moves as plausible as the reverse-stress scenario, the dashed line every move that loses exactly \$6.4m, and the scenario is where the two touch; any other point on the line, including the hand-written one, lies outside the ellipse, where the density is lower.

Stress tests and probability-based measures are used side by side rather than combined. A typical limit framework sets a VaR or ES limit and, separately, a ceiling on the loss under each of a handful of named scenarios; a breach of either requires action. Some firms attach rough probabilities to their scenarios to fold them into a single number, but that reintroduces

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

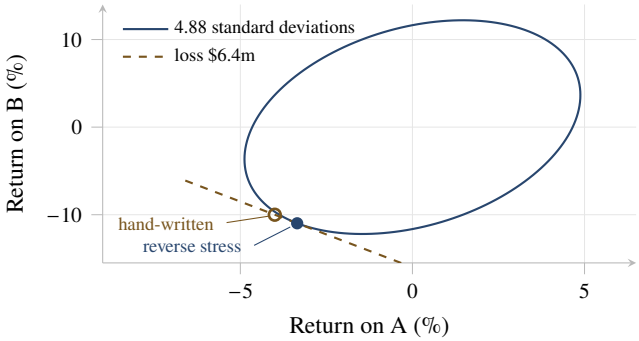


Figure 38.8: The most plausible move that loses $6.4m on the book of Example 38.14: the point where the iso-loss line touches a contour of the return distribution.

the estimate the stress test was built to avoid. The next section turns to the regulatory use of a crisis window, which keeps the probability machinery and changes only its data.

38.18 STRESSED MEASURES AND PROCYCLICALITY

Every estimator in this chapter is fitted to a window, and a window shrinks the measure as markets calm. Take a $100m position at 99%, one day: at a calm 1.0% daily volatility the VaR is $2.33m; estimated on a stressed window at 3.2%, it is $7.44m. The same book comes out a factor of 3.2 apart, and the difference lies entirely in the sample.

The consequence is perverse and it is structural. Risk-based capital falls through a calm period, permitting more risk to be taken, and rises after volatility arrives, forcing positions to be cut into a falling market. The measure amplifies the cycle it is meant to guard against—which the reader has already met as Section 8.9's collateral loop, appearing here as a capital spiral.

The repair is to add a measure that does not depend on the current window. A **stressed VaR**, added by Basel 2.5 in 2009, runs the same portfolio through a fixed twelve-month period of significant stress, so a bank's requirement cannot fall simply because the recent past was quiet. It is a crude device and it works, and Chapter 39 gives the current form.

What it does not fix is that any window, stressed or not, is history. A stressed window contains the crisis that happened, so it protects against the last crisis rather than the next one. And books are optimized against the measure, which means the positions that look cheap under it are precisely the ones whose risk it does not see.

38.19 WHAT THESE MEASURES DO NOT SEE

A risk measure describes a distribution, and not everything that matters about a book is in it.

Liquidity is absent. The measure assumes the position can be exited at the marked price; the gilt funds of Section 8.9, selling into a market their own sales were moving, could not.

Concentration is invisible. Two books with the same distribution have the same measure, whether the position is 1% of the market or 40% of it—and Amaranth, Archegos and the nickel short in Chapter 40 are all of the second kind.

Funding is a separate risk. A position can be correct, solvent, and still fail because variation margin must be paid before the offsetting gain is realized. Example 37.2's basis trade loses nothing and can still be closed for want of cash, and Chapter 40's Metallgesellschaft and LDI cases are the same mechanism at scale.

And the model itself is a risk. Every number here is conditional on the mapping from market factors to portfolio value being right, the mapping that Chapter 21's calibration and,

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

increasingly, trained networks produce. Chapter 39 treats that as its own subject, because it is the one exposure the risk system cannot measure with a risk system.

Case Study 38.1 Twenty-five sigma, several days running

In August 2007 a number of quantitative equity funds lost heavily over the same few days. The chief financial officer of Goldman Sachs, asked to explain, said the firm's funds had been seeing twenty-five standard deviation moves, several days in a row.

Take the statement literally and it is not a description of an event. Under a normal distribution a twenty-five sigma move has a probability of roughly 10^{-138}. There have been about 10^4 trading days in a working lifetime and something like 10^{17} seconds since the universe began. Several such days in one week is not a rare event; it is a refutation of the distribution used to measure it.

So the sentence has two readings and only one survives. Either something astronomically improbable happened repeatedly, or the model assigning the probabilities was wrong. Section 38.8 has already measured how wrong a normal assumption can be in the tail, and this is that error stated by someone who had just paid for it.

What the measure could not see was crowding. The positions were not unusual individually. They were held, in similar amounts and the same direction, by many funds running similar signals, so the deleveraging of one marked the book of the next. The exposure was to other people's positions, and a measure calibrated on a firm's own history contains no variable for that, which is why concentration belongs in this section rather than among the estimators.

Backtesting could not have raised the alarm in time. Example 38.13 showed that a year of exceptions cannot reject a 99% model whose realized coverage looks like 97.6%, and that even a model whose true coverage is 97% stays in the green zone one year in eight. A book can run on a badly shaped distribution for a long while and produce an exception count no supervisor would question.

The practical rule is the one the episode states out loud. Backtesting checks the count; the test that matters is not whether the count of breaches looks reasonable but whether the shape is right. A model that reports twenty-five sigma has already told you it is the wrong model; no further evidence is required, and none should be waited for.

38.20 WHO USES THESE NUMBERS, AND WHY

Regulators use them to set capital, the use that decides the definitions: Chapter 39's rules are the reason the industry computes expected shortfall at 97.5% rather than VaR's 99%.

Boards and senior management use VaR because it is one number with a probability attached, and that is genuinely the reason it won: the measure that can be explained in a sentence displaced measures that could not, and its statistical defects were formalized only later.

Risk managers use component and marginal figures rather than totals, because the total is not actionable and the breakdown in Example 38.12 is.

Clearing houses and prime brokers use short-horizon measures to set margin, where the horizon is a close-out estimate and the confidence level is a business decision about how often to be wrong. The large swap clearing houses compute initial margin by volatility-scaled historical simulation, which the industry often calls filtered historical simulation, taking a VaR or an expected shortfall over a lookback of several years that includes a period of stress, so that a quiet year does not shrink the margin. LCH's SwapClear, the largest of them, uses expected shortfall at 99.7% on volatility-scaled scenarios from a ten-year lookback period, with a five-day close-out for house accounts.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Insurers use the same idea at a different confidence and horizon. Solvency II, the European Union's insurance regime, sets the solvency capital requirement at the 99.5% VaR of an insurer's basic own funds, its assets less its liabilities, over one year, and an insurer with an approved internal model computes it by simulation.

Asset managers meet VaR as a limit on their funds. A European fund authorized under the rules for Undertakings for Collective Investment in Transferable Securities (UCITS) that measures its exposure by VaR must keep its 99%, twenty-day VaR below 20% of net asset value, or below twice that of a reference portfolio. In the United States, the SEC's Rule 18f-4, with which funds have had to comply since August 2022, imposes the same pair of limits on an open-end fund whose derivatives exposure exceeds 10% of its net assets.

Nonfinancial companies seldom manage by VaR, though some large ones disclose it, because their exposure is to operating cash flow rather than to the market value of a book. The variant they use is **cash-flow-at-risk**: the shortfall of a quarter's or a year's operating cash flow below its forecast that is exceeded with a stated probability.

Traders use them mainly as constraints, and a constraint is a thing to optimize against—which is why a measure's blind spots become positions. The London Whale of Section 40.10 is the plainest case: the measure changed and the book grew into it.

38.21 WHERE THIS CAME FROM

Value at risk was a practitioner construction that regulation adopted, and the theory arrived afterward to tell it what was wrong. Artzner, Delbaen, Eber and Heath (1999) sets out the axioms a risk measure should satisfy, shows that VaR violates subadditivity, and thereby names a defect desks had already met without being able to explain. Expected shortfall in its integral form satisfies them, as Acerbi and Tasche (2002) and Rockafellar and Uryasev (2002) showed for distributions with atoms, and the Basel framework moved to it in 2016, seventeen years after Artzner and his coauthors, an unusually clean case of a theoretical result changing a rule.

Embrechts, Klüppelberg and Mikosch (1997) is the extreme-value treatment behind Section 38.9's tail estimation, and McNeil, Frey and Embrechts (2015) is the modern synthesis.

SUMMARY

A risk measure turns a distribution into a number, and the choice of number is the whole subject. VaR is the quantile of (38.1): it says where the tail starts and nothing about what is in it, while expected shortfall averages the tail beyond that point. The Basel move from 99% VaR to 97.5% ES was calibrated to be capital-neutral under normality, so what changed was not the level but the sensitivity to tail shape.

That sensitivity is the case for ES. VaR can fail subadditivity where losses jump near the quantile: two ordinary bonds, each with a VaR of zero, can have a large VaR together. ES is coherent. The sharper argument is shape: a fat-tailed fit with the same volatility can lower VaR while raising ES, so the normal assumption understates exactly the tail ES measures.

Judgment enters through the inputs. EWMA is GARCH without a long-run level, so only GARCH produces a volatility term structure. Correlation is the weak input, and it moves most in the episodes the measure exists to describe; and correlation is not the whole of dependence, since a t copula makes joint extremes far more likely than a Gaussian one with the same correlation. Factor adequacy is a property of the portfolio, not of the curve, and allocation needs three figures, component, marginal and incremental VaR, because it asks three questions.

Estimation is where most of the error lives. Historical simulation is only as good as its window, and no reweighting adds information about a tail the window never saw; extreme value theory reaches beyond the data at a price in variance, and option books need a nonlinear

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

treatment. Stress testing prices a named scenario with no probability attached, and reverse stress testing runs backward from a loss to the most plausible scenario that produces it.

Validation is weaker than anyone would like. At 99% over 250 days a wide range of exception counts is consistent with a correct model, and expected shortfall cannot be backtested by counting at all, which is why Chapter 39 backtests the VaR and capitalizes the ES. Because every estimator is fitted to a window, the requirement is smallest just before it is needed, which stressed measures exist to correct. Liquidity, concentration, funding and the model's own correctness lie outside any risk measure; Chapter 39 prices some and Chapter 40 shows the rest.

FURTHER READING

The coherence axioms and the subadditivity failure of Example 38.2 come from Artzner, Delbaen, Eber and Heath (1999), which is short and states the counterexample plainly; Acerbi and Tasche (2002) settles why the integral form of expected shortfall, and not the conditional mean, is the coherent one. On estimation, J.P. Morgan's RiskMetrics technical document (1996), which made EWMA the industry's default, states why $\lambda = 0.94$ and what it costs, and the GARCH literature, which begins with Bollerslev (1986), is best approached through its forecasting results rather than its estimation theory; Example 38.8's term structure is the part that matters for derivatives. On copulas, read Section 26.6's sources first: the tail-dependence result of Example 38.10 was known well before it was needed and ignored anyway, so read them for the history rather than the mathematics. On backtesting, Kupiec (1995) and the Basel Committee's 1996 backtesting framework are worth reading side by side precisely because they disagree, and the supervisory document sets out how its zones trade false alarms against the power to catch a bad model. Christoffersen (1998) is the paper for the other half of the question: it is short, and it is where the independence test of Section 38.16 comes from, together with the observation that coverage and independence are separable. Hull and White (1998) is the short paper behind volatility scaling, and Barone-Adesi, Giannopoulos and Vosper (1999) the one behind filtered historical simulation. On backtesting expected shortfall, Acerbi and Székely (2014) is the place to start, and Gneiting (2011) and Fissler and Ziegel (2016) supply the theory of why ES cannot be scored alone and how the pair with VaR can be. Chapter 39 gives the sources for the current capital framework, and [TN38] derives the representation that makes expected shortfall optimizable.

PRACTICE QUESTIONS

38.1 Using Example 38.1, state the 99% one-day VaR and ES for a \$100m book at 2% daily volatility, and explain why the ratio between them falls as confidence rises.

38.2 Show that 99% VaR and 97.5% ES nearly coincide for a normal loss, and explain what the regulator was trying to achieve by choosing those two levels.

38.3 State the subadditivity condition and explain why a measure that violates it cannot be used for desk limits.

38.4 Using Example 38.2, compute the 99% VaR of each bond alone and of the two together, and explain what feature of the loss distribution produces the failure.

38.5 Compute the 99% expected shortfall for the same two bonds, and verify that it satisfies subadditivity.

38.6 Explain why a desk measured on VaR has an incentive to sell far-out-of-the-money options, and what expected shortfall does to that incentive.

38.7 Explain what historical simulation assumes, and state the largest loss it is capable of reporting.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

38.8 Using Example 38.6, explain why the fat-tailed VaR at 95% is *lower* than the normal VaR, and why the expected shortfall is not.

38.9 Compute, from Example 38.6, the ratio of the fat-tailed to the normal expected shortfall at 99.9% and the same ratio for VaR, and explain why the expected shortfall error matters more than the VaR error.

38.10 Using the simulation that follows Example 38.7, compare the bias and the variability of the three estimators, and explain why the most precise one is the most wrong.

38.11 Compute the probability that a 500-day window's largest loss falls below the true 99.9% VaR, and explain what this means for historical simulation at that confidence.

38.12 Give the two reasons extreme value theory converts an impossible estimate into a difficult one rather than into a known answer.

38.13 Using Example 38.8, compute the long-run volatility implied by $\omega = 0.000002$, $\alpha = 0.06$, $\beta = 0.93$, and state the one-year forecast under each estimator.

38.14 Explain why EWMA is adequate for a one-day margin calculation and inadequate for a volatility term structure.

38.15 Using (38.4), recompute the VaR of Example 38.9's book at correlations of 0.30 and 0.90, and explain why the estimation error has a consistent sign.

38.16 Using Example 38.10, compute the probability that the second exposure is beyond its 99.9th percentile given that the first is, under each copula, and compare the two with the coefficient of tail dependence.

38.17 Using Example 38.11, explain why three factors keep 99% of the directional book's variance and 10% of the butterfly's, and state what a risk system must report as a result.

38.18 State the square-root-of-time rule, the assumptions it requires, and the direction of its error in a trending market.

38.19 Using Example 38.12, state each position's share of the money and share of the risk, and explain what a total-VaR limit fails to notice.

38.20 Explain how an allocation can be exact while the total it allocates is unreliable.

38.21 Using Example 38.13, state the range of exception counts consistent with a correct model at 99% over 250 days, and explain why zero exceptions is also a rejection.

38.22 Explain where the Basel zones and the Kupiec test disagree, and what the supervisor gains by setting the zones where they are.

38.23 Explain why expected shortfall cannot be backtested by counting exceptions, and how the capital framework works around this.

38.24 Using Examples 38.14 and 38.15, explain what a stress test reports that VaR does not, why a one-factor-at-a-time sensitivity report understated the scenario loss, and what the reverse stress test found about where the book's risk lies.

38.25 A book's daily loss is normal with a standard deviation of \$2 million. Using the standard-error formula of Section 38.5, compute the standard error of a 99% VaR estimated from 250, 500 and 1,000 days, each as a share of the true VaR, and state how much longer a window must be to halve it.

38.26 Repeat Example 38.5 with a dollar gamma of −\$200 million. Compute the 99% one-day loss under the delta–gamma expansion, and state by how much it exceeds the delta-only figure.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

38.27 Using the GARCH parameters of Example 38.8, show that a deviation of the variance from its long-run level halves in about 69 trading days, and compute the annualized volatility forecast for the single day 100 trading days after the variance just estimated.

38.28 Using (38.8), compute the ten-day 99% VaR of the one-day \$4.65m book of Section 38.14 at a daily autocorrelation of -0.10, and state the size and direction of the square-root rule's error.

38.29 Redo Example 38.12 at a correlation of 0.60: compute the portfolio VaR, each position's marginal, component and incremental VaR, and state which of the three sets sums to the total.

38.30 For the two-stock book of Example 38.4, find the most plausible pair of returns that loses \$1.0 million, state how many standard deviations out it lies, and compare it with the scenario in which both stocks fall by four of their own standard deviations.

FURTHER QUESTIONS

38.31 Example 38.2 constructs a subadditivity failure with two bonds. Determine the conditions on default probability, recovery, and confidence level under which the failure occurs, state how the region changes as the number of bonds grows, and explain what this implies for a real credit portfolio measured at 99%.

38.32 Example 38.6 shows the fat-tailed VaR falling below the normal VaR at 95%. Find the confidence level at which the two cross for a t distribution with ν degrees of freedom scaled to a fixed variance, determine how that crossing point moves with ν, and state what it implies for a firm that measures at 95% and fits fat tails.

38.33 Example 38.8 compares forecast term structures. Fit both estimators to a real equity index series, compare their forecasts against subsequently realized volatility at horizons from one day to one year, and determine the horizon beyond which the mean reversion earns its extra parameter.

38.34 Section 38.12 shows that a Gaussian copula has zero tail dependence. Construct a portfolio of credit exposures whose 99.9% expected shortfall is materially different under Gaussian and t copulas calibrated to the same pairwise correlations, and determine what data would be needed to choose between them.

38.35 Section 38.14 treats the horizon as a property of the position rather than of the firm. Design a liquidity-horizon scheme for a book containing listed futures, cleared swaps, and bespoke structured trades; state how you would estimate each horizon; and determine how much the aggregate measure changes against a uniform ten-day assumption.

38.36 Section 38.16 notes that expected shortfall resists backtesting. Survey the proposed tests, implement one, and determine how many observations it needs to distinguish a correct model from one miscalibrated by 20%. Then state whether that number is available in practice.

38.37 Section 38.17 solves reverse stress testing in closed form for a linear book. Take the book of Example 38.5, the sold puts on stock 1 beside the \$5 million in stock 2, find numerically the most plausible pair of returns that loses \$1.0 million under the delta–gamma expansion, compare it with the linear answer, and explain why the gamma term moves the scenario.

38.38 Section 38.18 argues that books are optimized against whatever is measured. Take a measure from this chapter, construct the portfolio that maximizes true risk subject to a fixed reported value of it, and state what supervisory rule would prevent the construction.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Chapter 39
Market Risk Capital and Model Risk

Chapter 38 built measures of what a book can lose. This chapter is about what a regulator does with them, and about the exposure that no measure in Chapter 38 can see: the risk that the model producing the numbers is wrong.

The two subjects belong together for a reason that becomes obvious once stated. A capital rule is a contract with a bank about a number, and the number comes from a model the bank builds. So the rule must either specify the model or test it, and the modern framework does both—which is why so much of it is machinery for deciding whether a bank's own model may be used at all.

One caution comes before any of the detail. The implementation dates in this area have moved repeatedly and in every major jurisdiction. As of November 2026, they stood as follows:

- the Basel standard's own implementation date was January 1, 2023;
- Canada and Japan had applied it since late 2023 and March 2024 respectively;
- China had applied it since January 1, 2024;
- the European Union was due to apply its market risk rules from January 1, 2027, after two postponements, with a temporary multiplier, adopted in June 2026, offsetting the capital increase until the end of 2029;
- the United Kingdom was due to apply them from the same date, with its internal models approach for market risk deferred to January 1, 2028; the Prudential Regulation Authority (PRA) consulted on adjustments to it in June 2026;
- the United States had re-proposed its version in March 2026 with no final rule.

Check the calendar against the date you read this. The structure below is stable, and the calendar is not.

Sections 39.1 and 39.2 explain why the trading book has its own capital rule and where its boundary sits. Sections 39.3 to 39.8 are the rule itself: the standardized approach, internal models, non-modellable risk factors, the P&L attribution test, desk-level backtesting, and the approval cliff those tests create. Sections 39.9 to 39.12 assemble the total and cover the three channels that sit above or beside it: the output floor, prudent valuation, and stress testing. Section 39.13 asks what the framework changed and who bears the cost, and Sections 39.14 and 39.15 turn to model risk, including the validation of models that cannot be verified.

39.1 WHY THE TRADING BOOK NEEDS ITS OWN RULE

A bank keeps two books. The loans and bonds in the banking book are mostly held to maturity and capitalized against default. The positions in the trading book are held to be sold, so the loss that matters is a price move over a short horizon, not a default over years.

The framework that preceded the current one was built on Chapter 38's VaR and failed in 2008 in three identifiable ways. It used a ten-day 99% VaR calibrated on recent data, so it was

procyclical exactly as Section 38.18 describes. It capitalized the trading book far more lightly than the banking book for the same economic exposure, which produced a decade of moving assets across the boundary. And it assumed every position could be exited in ten days.

The current framework is the Basel Committee's **Fundamental Review of the Trading Book**, known by its initials, **FRTB**, which is the name to search for; the US implementation was folded into the package called the **Basel III endgame**, whose scope has been contested and revised since it was first proposed. It answers each of the three failures:

- *procyclicality*, by calibrating the internal-model measure to a period of stress rather than to recent data, so that it cannot fall because markets are calm (Section 39.4);
- *the light trading book*, by fixing which book a position belongs in with a prescriptive list, and by denying any capital benefit to moving it (Section 39.2);
- *liquidity*, by giving each risk factor a horizon of 10 to 120 days, so that a risk factor that takes longer to exit is charged over a longer horizon (Section 39.4).

It is worth reading that way: every complication below exists because something specific broke. The framework offers two ways to compute the charge, the **standardized approach**, which every bank computes, and the **internal models approach**, which a bank may use desk by desk once its models are approved.

Case Study 39.1 The patch, and why it was not enough

The response to 2008 came in two stages, and the gap between them is instructive.

The immediate patch arrived in July 2009 and is known as **Basel 2.5**. It added a stressed VaR computed on a crisis window, and an **incremental risk charge** for default and migration in the trading book. Both were added to the existing measure rather than replacing it, and the Basel Committee's (2009) impact study put the combined effect at two to three times existing trading-book capital.

That the patch worked is not in dispute; that it was a patch is the point. Stressed VaR addressed procyclicality by adding a second calculation rather than fixing the first, so banks ran two measures on the same book and held the sum. The incremental risk charge covered default in the trading book because the boundary had let credit exposures migrate there, treating the symptom rather than the boundary; it did use liquidity horizons, with a three-month floor, but only for default and migration. Nothing in the patch addressed liquidity in the VaR measures themselves: a position was still assumed exitable in ten days.

Of the three failures above, the patch had met only procyclicality, and only by adding a calculation. The boundary and the horizon problems survived it, and so did a defect in the measure itself; those three were left to the current framework. The boundary was still intent-based, until Section 39.2's prescriptive list. The horizon was still a constant ten days, until Example 39.3's schedule of liquidity horizons. And the measure was still VaR, with Section 38.4's subadditivity failure and Section 38.8's finding that a fatter tail can lower VaR intact. That is the direct reason the current framework is written around expected shortfall.

The general lesson is about how frameworks fail. The 2009 patch was fast, effective in raising capital, and left every structural defect in place. Its successor took a decade and is still being implemented: a fast patch or a slow cure is the choice a supervisor faces whenever the diagnosis is architectural and the pressure is immediate.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

39.2 THE BOUNDARY BETWEEN THE TRADING BOOK AND THE BANKING BOOK

The first fix concerns the line between the two books of Section 39.1: the trading book, capitalized against price moves under this chapter's rules, and the banking book, capitalized against default under the credit rules of Section 27.11 and Section 39.10. It is the least mathematical fix and was the most consequential. If capital differs between the two books, the boundary between them is a valuable option, and banks exercised it. A bank holding $500 million of corporate bonds that attracted 3% capital in the trading book and 8% in the banking book had a five-point capital difference riding on a classification, and under the old rule the classification turned on the bank's stated intent.

The current rule makes the boundary prescriptive rather than intent-based. A presumptive list says what sits where:

- instruments held for short-term resale, positions in the correlation trading portfolio (securitization tranches and *n*th-to-default credit derivatives), net short credit or equity positions, and anything arising from underwriting or market making are presumed to be in the trading book;
- unlisted equities, real estate, retail and small-business credit, and equity investments in funds whose holdings the bank cannot identify and value daily are in the banking book.

A bank may depart from the presumption only with supervisory approval and documented reasons. Reassignment between the books requires supervisory approval, is permitted only in extraordinary circumstances, and may not produce a capital benefit: if moving a position lowers its charge, the bank holds a disclosed surcharge equal to the difference. The option was not tightened; it was removed, which is the right response to an incentive rather than a modeling problem.

39.3 THE STANDARDIZED APPROACH

Every bank must compute the standardized charge, whether or not it has model approval, and the charge has three parts.

The sensitivities-based method is the bulk of it. A desk reports delta, vega, and curvature sensitivities to prescribed risk factors; the regulator supplies risk weights and correlations. The charge then combines them by a prescribed formula: each sensitivity is multiplied by its risk weight, and the products are combined the way a portfolio's standard deviation is built from the variance of equation (38.4) in Chapter 38's model-building approach, as the square root of their squares plus their correlated cross-products. Example 39.1 works it through. The three sensitivities are the coefficients of Chapter 18's expansion, equation (18.1), read as a capital rule: delta charges its first-order term, vega its volatility term, and curvature its second-order term. The correlations are given, not estimated. Section 38.11 showed that correlation is the least stable input to a risk measure: estimated in calm periods and applied in violent ones, so that a long book's measured risk comes out too low. The framework settles the question by fiat: the supervisor would rather set the number and charge for doubt about it than depend on a bank's estimate.

The default risk charge covers **jump-to-default**, the loss if an issuer defaults overnight with no warning in its spread, on traded bonds, credit derivatives and equities. **The residual risk add-on** is a flat charge on payoffs the sensitivities do not describe. The next three subsections take delta and vega, curvature, and the other two parts.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

39.3.1 Delta and vega

Sensitivities are grouped into **buckets** (one currency's interest rate curve, one equity sector) and aggregated in two stages, within each bucket and then across buckets. The delta charge is worth working through, because its structure contains a device that has no analog in Chapter 38 and that determines which books the framework treats harshly.

Example 39.1 Three correlation scenarios, and which one binds

Consider one currency and three points on its interest rate curve, at 2, 5 and 10 years. For each point k the bank reports its net sensitivity s_k, the loss on all its positions on this curve when the rate at that point alone rises by one basis point. This is Section 7.1's bucketed DV01, in dollars per basis point, so receiving and paying fixed at the same point carry opposite signs. The regulator supplies a risk weight RW_k for each point, here 130, 110 and 110 basis points (not the percentage of an exposure that Section 27.11's credit weights are, but the size of the rate move it prescribes as a shock), and the correlations

$$\rho_{kl} = \max\left(e^{-0.03\,|T_k - T_l|/\min(T_k, T_l)},\ 0.40\right),$$

where T_k is the tenor of point k in years. These give 0.9560 between 2 and 5 years, 0.9704 between 5 and 10 years, and 0.8869 between 2 and 10 years. Both the risk weights and this correlation are the Basel standard's own, not choices made for the example (for the major currencies a bank may divide the weights by $\sqrt{2}$). The correlation belongs to the family Chapter 32 used for forward rates, Example 32.1's exponential decay and the floored blend that follows it, with two differences: the distance is measured relative to the shorter tenor, so the short end decorrelates faster (2 and 5 years at 0.956, against 0.991 for the same three-year gap between 10 and 13 years), and the floor of 0.40 is a cutoff rather than a blend. The **weighted sensitivity** $WS_k = s_k \times RW_k$ is then the dollar loss from that prescribed move at point k, and the bucket charge is

$$K_b = \sqrt{\max\left(0,\ \sum_k WS_k^2 + \sum_{k \neq l} \rho_{kl}\, WS_k\, WS_l\right)},$$

where the floor at zero matters only when the correlations are moved, as they are next.

The charge is computed three times, once with the prescribed correlations and once with each of

$$\rho_{kl}^{\text{high}} = \min\left(1.25\,\rho_{kl},\ 1\right), \qquad \rho_{kl}^{\text{low}} = \max\left(2\rho_{kl} - 1,\ 0.75\,\rho_{kl}\right).$$

The capital number is the largest of the three charges, not the charge at the smallest correlations: which set of correlations binds depends on the book, as the two books below show. (Basel takes this maximum after the buckets have been aggregated; with a single bucket, as here, that is the same thing.)

Take two books. The first receives fixed at all three points, with sensitivities of \$2,000, \$3,000 and \$4,000 per basis point. The second is a butterfly, the kind of curve trade Example 38.11 measures, here with wings at 2 and 10 years and the body at 5: it receives fixed at 2 and 10 years (\$2,200 and \$2,000 per basis point) and pays fixed at 5 years (\$4,600), sized so that its weighted sensitivities sum to zero: the prescribed shocks, 130 basis points at 2 years and 110 at 5 and 10, leave it unchanged, although a parallel one-basis-point move changes its value by \$400.

For the directional book the weighted sensitivities are \$260,000, \$330,000 and \$440,000, so under the prescribed correlations

$$\begin{aligned} K_b^2 &= 260^2 + 330^2 + 440^2 + 2 \times (0.9560 \times 260 \times 330 \\ &\qquad + 0.8869 \times 260 \times 440 + 0.9704 \times 330 \times 440) = 1{,}018{,}894, \end{aligned}$$

in thousands of dollars squared and with the correlations unrounded, so K_b = \$1,009,403. The three scenarios give:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Scenario	Directional book	Butterfly
Low correlation	\$988,376	**\$100,854**
Medium (as prescribed)	\$1,009,403	\$71,315
High correlation	**\$1,030,000**	\$0
Charge (worst)	**\$1,030,000**	**\$100,854**
Excess over medium	+2.0%	+41.4%

The two books are punished by opposite scenarios. The directional book gets worse as the curve moves together, so *high* correlation binds, and the penalty is mild, 2.0%, because a directional book is nearly as risky under any correlation. The butterfly is the reverse: under high correlation its charge falls to *exactly zero*, since a perfectly correlated curve makes the hedge perfect, while under low correlation the legs stop offsetting and the charge is 41.4% above the prescribed case.

So the hedged book pays the larger proportional penalty, and the reason is the design working as intended: the rule refuses to let a bank choose the correlation that flatters its book. A framework that used a single prescribed correlation could be arbitraged by constructing positions that are riskless under it: a book hedged only at the prescribed number is unhedged at any other. Taking the worst of three removes that possibility.

The cost is borne by genuine relative-value business. A butterfly is a real hedge whose risk really is small when the curve moves together, and the framework charges it as though the correlations it relies on might not hold. Section 38.11 showed that correlations fail in exactly the episodes capital exists for, so the charge is defensible even though it is expensive.

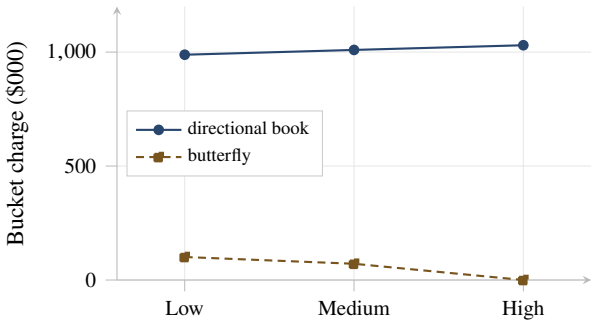


Figure 39.1: Bucket charge under the three correlation scenarios, for the two books of Example 39.1.

Figure 39.1 shows why one rule serves both books. The two lines slope in opposite directions, so no single correlation is conservative for both—and any rule that picked one would leave the other exploitable. Evaluating three scenarios and taking the worst is a simple way to be conservative against a quantity no one can estimate reliably.

Vega is aggregated by the same two-stage formula, with correlations of its own ([TN39] works a book of options through it). Its risk weight is not a fixed number but $\min\left(55\% \times \sqrt{\mathrm{LH}/10},\ 100\%\right)$, where LH is the liquidity horizon the framework assigns to the volatility, in days: 20 days for the volatility of a large-cap stock, giving 77.8%, and 40 days or more for every other volatility, giving 100%.

39.3.2 Curvature

Curvature is charged separately because delta and vega miss the convexity that Chapters 34 and 35 showed dominates structured books. The charge, CVR, shocks the factor up and down by its delta risk weight, revalues in full, and asks how much worse the result is than the delta

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

prediction:

$$\text{CVR} = \max\Big(0,\ -\min_{\pm}\big[V\big(S(1 \pm RW)\big) - V(S) \mp RW \times \Delta_{\$}\big]\Big).$$

Here $V(S)$ is the position's value at the stock price S, $\Delta_{\$}$ its dollar delta as in Chapter 38, and RW the delta risk weight. It revalues rather than multiplying a reported gamma by $\frac{1}{2}(\Delta S)^2$ because the shocks are large, and Section 18.10 showed that the expansion's gamma term misstates a large move. The charge has an asymmetry that surprises people: convexity decides its sign, and the floor at zero turns that sign into all or nothing. (Basel applies that floor when it aggregates the risk factors; for a single factor, as here, the effect is the same.)

Take a long \$10 million notional at-the-money six-month call on a large-cap emerging-market stock, at 25% volatility with interest rates at zero. At the money $d_1 = 0.25\sqrt{0.5}/2 = 0.0884$, so the option is worth $\$10\text{m} \times [N(0.0884) - N(-0.0884)] = \$704{,}000$ and its delta is $N(0.0884) = 0.5352$, a dollar delta of \$5.352m. The stock's delta risk weight, and so its curvature shock, is 55%, a Basel weight for a large-cap emerging-market stock (45% to 60% by sector there, 30% to 50% in advanced economies). Up 55%, the option is worth \$5.505m against a delta prediction of \$0.704m + 0.55 × \$5.352m = \$3.648m; down 55%, it is worth almost nothing against a prediction of −\$2.239m. Revaluation beats the delta prediction by \$1.86m on the upside and \$2.24m on the downside, both favorable, because the holder is long gamma. The bracketed term is positive in both directions, so its minimum is positive, and the floor sets the charge to zero. Sell the same option and both figures change sign, giving a charge of **\$2.24m**. Figure 39.2 draws the two gaps.

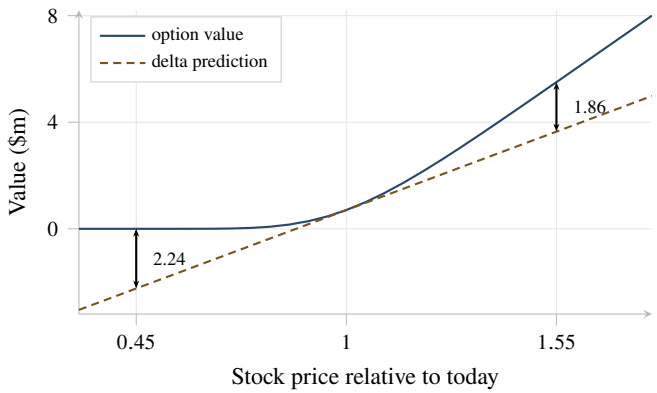


Figure 39.2: A long call's value and its delta prediction, with the gaps at the ±55% curvature shocks. Illustrative.

Both gaps lie on the same side of the line because the value curve is convex; the floor then removes the long holder's negative charge. Long optionality attracts no curvature charge and short optionality attracts all of it, which is the correct treatment and worth stating plainly because it is the first place in this book where a capital rule agrees exactly with Chapter 18's economics. The vega charge completes the picture, and the three components together answer a question a trading desk will ask immediately: does hedging reduce the charge?

Example 39.2 A delta-hedged short option position

Sell the \$10 million notional six-month at-the-money call above, on the same large-cap emerging-market stock at 25% volatility, and delta-hedge it.

Vega is reported as the sensitivity to a *relative* move in implied volatility, the desk's vega multiplied by the volatility itself, so the option's vega of $\$10\text{m} \times \sqrt{0.5}\, N'(0.0884)/100 = \$28{,}099.50$

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

per volatility point is reported as $28{,}099.50 \times 25 = \$702{,}488$. The volatility of a large-cap stock has a 20-day liquidity horizon, so its vega risk weight is $55\% \times \sqrt{2} = 77.8\%$ and the vega charge is $\$702{,}488 \times 55\% \times \sqrt{2} = \$546{,}407$.

Component	Charge	Why
Delta	≈ \$0	removed by the hedge
Vega	\$546,407	the hedge does not touch it
Curvature	\$2,239,369	short gamma, both directions
Total	**\$2,785,776**	**27.9% of notional**

Delta-hedging removed the delta charge and none of the rest. A book that is flat by the measure a trader watches minute to minute attracts capital equal to more than a quarter of its notional, and 80% of that comes from the curvature term alone.

This is the framework doing deliberately what Chapter 18 says is true. Delta neutrality is a statement about the first derivative at a point; the risk in a short option is the second derivative and the volatility, and a capital rule that let delta-hedging extinguish the charge would be paying for a hedge that does not hedge the exposure.

It also explains a pricing fact from Chapter 35. An issuer selling optionality to retail carries a charge of this kind for as long as it keeps the risk, on a delta-hedged book, so the capital cost is a running cost rather than a one-off, and Section 35.10's fee arithmetic has to cover it before anything is left over.

39.3.3 The default risk charge and the residual risk add-on

The default risk charge treats default as an event the sensitivities cannot see: a bond's spread sensitivity says what a widening costs, not what a default costs. Each position's jump-to-default loss is its loss given default on the notional, less any loss its mark already shows. Longs and shorts in the same issuer offset, offsets between different issuers in the same bucket are recognized only in part, and the net amount is multiplied by a risk weight set by credit quality. A desk long a bond and holding bought protection on the same issuer pays little; a desk long one issuer and short an equal amount of another of the same rating in the same bucket still pays half the long position's charge. The charge is calibrated as the banking book's credit capital is, to a one-year horizon at 99.9%, Section 24.14's worst-case default rate, so that a bond's default capital is broadly comparable whichever book holds it; a bank on internal models computes it by simulating defaults over that year with at least two systematic factors. Migration is left out on purpose, because the spread sensitivities already measure what a downgrade costs.

The residual risk add-on is the framework's admission that some payoffs are beyond its sensitivities. It is 1% of the notional of instruments with an exotic underlying—weather, longevity, future realized volatility (a variance swap's underlying)—and 0.1% of the notional of other instruments bearing residual risks, such as gap risk, correlation risk or behavioral risk (a borrower's option to prepay, for instance). A barrier or a correlation product attracts the add-on because the framework declines to model what its sensitivities miss, which is more honest than a false precision would be.

39.4 THE INTERNAL MODELS APPROACH

A bank with approval computes expected shortfall at 97.5%, Chapter 38's measure at the confidence level Example 38.1 explained, calibrated to a period of stress rather than to recent data. The resulting charge, the **internally modeled capital charge** (IMCC), weights equally the expected shortfall of the whole book, with diversification across risk classes, and the sum

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

of the expected shortfalls of each risk class computed alone:

$$\text{IMCC} = \frac{1}{2}\,\text{ES}_{\text{all}} + \frac{1}{2}\sum_{c}\text{ES}_{c}.$$

Here ES_{all} shocks every risk factor together and ES_c only those of risk class c, so the bank's own correlations across risk classes enter with half weight and those within each class in full.

Two design choices follow from Chapter 38's failures. Stress calibration answers Section 38.18's procyclicality directly: the measure cannot fall because markets are calm. And the horizon is no longer a constant.

Example 39.3 The horizon schedule is a capital multiplier

Risk factors are assigned **liquidity horizons** of 10, 20, 40, 60, or 120 days by class:

Horizon	Examples
10 days	major-currency interest rates, large-cap equity prices, major exchange rates
20 days	other interest rates, investment-grade sovereign spreads, small-cap equity prices, other exchange rates, large-cap equity volatility, energy and metal prices
40 days	investment-grade corporate spreads, other sovereign spreads, exchange rate volatility
60 days	high-yield corporate spreads, interest rate volatility, other equity volatility, energy and metal volatility, other commodity prices
120 days	other credit spreads, credit spread volatility, other commodity volatility

Each expected shortfall in the IMCC is computed five times at a ten-day horizon: ES_1 shocks every factor, and ES_j shocks only the factors whose horizon is at least LH_j, the jth horizon in the schedule. The results are aggregated as

$$\text{ES} = \sqrt{\text{ES}_1^2 + \sum_{j\geq 2}\left[\text{ES}_j\sqrt{\frac{\text{LH}_j - \text{LH}_{j-1}}{10}}\right]^2},$$

with weights $\sqrt{(\text{LH}_j - \text{LH}_{j-1})/10}$ of 1.0000, 1.0000, 1.4142, 1.4142 and 2.4495 (taking $\text{LH}_0 = 0$, so that ES_1's weight is one). A factor with a 120-day horizon appears in all five terms, so its contribution is scaled by $\sqrt{1+1+2+2+6} = 3.46$ in total.

Take a desk whose ES values, for the factors surviving at each horizon, run 100, 90, 70, 45, 20. Then

$$\text{ES} = \sqrt{100^2 + 90^2 + (70\times 1.4142)^2 + (45\times 1.4142)^2 + (20\times 2.4495)^2} = 185.34$$

against a flat ten-day figure of 100. The desk's capital is 1.85 times the single-horizon figure for identical positions.

The weights are the square-root-of-time rule of Section 38.14, applied to the stretch of horizon each step adds. Section 38.14 showed that rule failing when returns are autocorrelated, and the framework uses it anyway, because it needs a formula a supervisor can prescribe. The horizon belongs to what is held, not to the rule, so the framework replaces the old single ten-day horizon with five, assigned by risk-factor class, though it keeps that horizon's square-root scaling, and the price is that a bank's capital now depends on a regulatory classification of its risk factors rather than on anything it measures. *The judgment moved rather than disappeared*, which is the recurring shape of this chapter.

39.5 NON-MODELLABLE RISK FACTORS

A factor may only be modeled if there is enough evidence of real prices for it. The **risk factor eligibility test** asks for at least 24 real price observations over the past twelve months, with no 90-day period containing fewer than four, or at least 100 observations over the twelve months. Anything failing that test is a **non-modellable risk factor** (NMRF) and is capitalized by a

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

stressed expected shortfall (SES): the loss when that factor alone receives its own stress shock, at least as severe as a 97.5% expected shortfall measured on that factor's own worst stress period, over its liquidity horizon or 20 days if longer.

The penalty is not only that the stress is severe. The non-modellable capital is also added to the modeled charge rather than diversified with it: the modellable part gets the benefit of Chapter 38's covariance structure, and the non-modellable part does not. Within the stress add-on itself, idiosyncratic credit and equity factors are combined as if independent and the rest by a formula with a parameter of 0.6 that amounts to a correlation of 0.36 between any two of them, so some diversification survives there, but none survives between the two parts.

Example 39.4 Two penalties for one illiquid factor

A desk's risk splits into a modellable part with a stand-alone expected shortfall of 80 and an illiquid factor with a stand-alone expected shortfall of 30, correlated at 0.25. If every factor were modellable, the desk would hold

$$\sqrt{80^2 + 30^2 + 2 \times 0.25 \times 80 \times 30} = 92.2.$$

The illiquid factor fails the eligibility test. Its stress scenario, calibrated over its own longer horizon, is assumed here to be 1.6 times its expected shortfall, so its add-on is $1.6 \times 30 = 48$, and the desk holds $80 + 48 = 128$, 38.8% more for identical positions.

The 35.8 difference has two sources of almost equal size. Adding rather than diversifying costs $80 + 30 - 92.2 = 17.8$, and the harsher stress calibration costs $48 - 30 = 18.0$. Only the first is a property of the rule; the 1.6 is this example's assumption.

So identical risk costs more when it sits in an illiquid factor, and that is deliberate: it prices illiquidity, which Section 38.19 put first among what the measure cannot see. The framework cannot model liquidity, so it taxes illiquidity instead—a substitution worth noticing, because the tax falls on the bespoke structured business of Chapter 35.

39.6 THE P&L ATTRIBUTION TEST

The attribution test decides whether a desk keeps its model, and it is the most quantitatively interesting thing in the framework. Approval is granted desk by desk, and a **trading desk** here is a unit the bank designates and the supervisor accepts: a group of traders or accounts running a well-defined business strategy, with its own risk management and a single head.

Each desk produces two daily numbers. **Hypothetical P&L** is the day's profit on yesterday's positions, computed by full front-office revaluation. **Risk-theoretical P&L** is the same thing computed using only the risk factors the risk model contains. If the risk model captures the desk, the two series agree.

The test compares them over 250 days with two statistics. The Spearman rank correlation asks whether the two series rise and fall on the same days. The **Kolmogorov–Smirnov** (KS) statistic is the largest gap between the two series' empirical distribution functions, and asks whether they have the same distribution. A desk is green if the rank correlation is above 0.80 and KS below 0.09, red if the rank correlation is below 0.70 or KS above 0.12, and amber otherwise.

Example 39.5 A third of the P&L may be unexplained

Let a fraction f of the desk's P&L variance come from factors the risk model does not contain, independent of the rest. The risk model revalues the positions with only the factors it contains, so its risk-theoretical series is the hypothetical series without that part: it has variance $1 - f$ of the hypothetical one, and the two correlate at $\rho = \sqrt{1 - f}$. For jointly normal series the rank

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

correlation is

$$\rho_S = \frac{6}{\pi}\arcsin\left(\frac{\rho}{2}\right).$$

KS has no closed form here, so the table gives its mean over 2,000 simulated 250-day samples:

Unexplained variance	ρ	Spearman	KS (mean)	Zone
5%	0.9747	0.9722	0.043	green
20%	0.8944	0.8855	0.065	green
30%	0.8367	0.8243	0.081	green
33.8%	0.8135	0.8000	0.088	**amber begins**
40%	0.7746	0.7595	0.099	amber
48.6%	0.7167	0.7000	0.117	**red begins**
50%	0.7071	0.6902	0.120	red

A desk can leave a third of its profit-and-loss variance unexplained by its own risk model and remain in the green zone. It takes half before the model is disqualified.

The two statistics leave green within a couple of percentage points of each other (the rank correlation at 33.8%, the mean KS at about 35%) and reach red together near half, because a missing factor does two things at once. It weakens the day-by-day correlation, which the rank correlation sees, and it shrinks the risk-theoretical series' variance, which KS sees. A model error that did only one of the two would be caught by only one statistic.

The KS column is a mean, and a single desk's KS is noisy. At 30% unexplained variance, where the mean is 0.081, one 250-day sample in four exceeds 0.09, so an otherwise green desk can be pushed into amber by the sample alone.

The practical reading is not that the test is useless. It catches a desk whose risk model misses a factor carrying half its variance, which is what it was built for. What it does not catch is a smaller omission: a missing factor carrying a quarter of the variance, or a model that tracks the hypothetical P&L closely while sitting slightly off it every day.

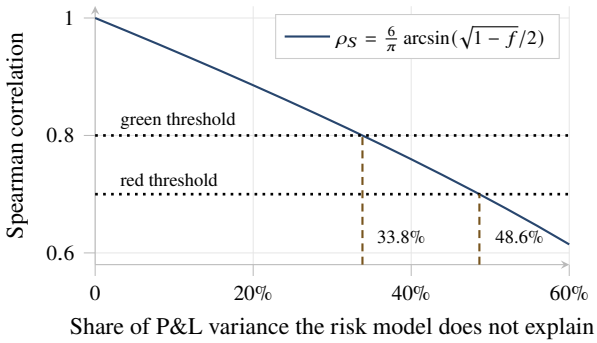


Figure 39.3: Rank correlation against unexplained P&L variance, with the zone thresholds.

Figure 39.3 is worth looking at before designing a risk model. The curve falls steadily, but the green threshold sits low on it: the rank correlation does not fall to 0.80 until a third of the variance is unexplained. The test is permissive because of where its threshold is set, and the choice of statistic adds a second weakness: rank correlation was chosen because it is robust to outliers, and robustness to outliers is robustness to the days that matter.

39.7 BACKTESTING AT DESK LEVEL

The attribution test is not the only hurdle, and the two interact. Each desk also backtests: Chapter 38's exception counting, run at both 97.5% and 99% on the desk's own one-day VaR.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

A desk with more than 12 exceptions at 99%, or more than 30 at 97.5%, over the past 250 days (2.5 and 6.25 on average for a correct model) loses its model approval and moves to the standardized approach.

VaR is backtested even though capital is held against expected shortfall, for the reason Section 38.16 gave: exceptions can be counted, and expected shortfall offers no event to count. The framework checks the quantile and trusts the tail to come from the same model.

Separately, the bank's model as a whole is backtested at 99% across all its approved desks, and that count sets the multiplier of Section 39.9. The desk-level test decides whether a desk may use its model; the bank-level test decides how much the modeled number is scaled up.

The two desk tests catch different faults, which is the point of having both. Backtesting asks whether the measure is *calibrated*: are losses exceeding it at the stated rate? Attribution asks whether the model *contains the right factors*: a factor the model lacks moves the hypothetical P&L but not the risk-theoretical P&L, so the two daily series stop tracking. A desk can pass either and fail the other.

A desk with a correctly sized but structurally wrong model passes backtesting and fails attribution once the missing factor is large. Its VaR is the right magnitude because its sensitivities to the factors it has were scaled up to the desk's total P&L variance, so the risk-theoretical series has the right variance and KS sees little. But the daily series do not track, and the rank correlation alone takes the desk to amber beyond about a third of the variance and to red near half (Example 39.5). A desk with the right factors and a stale calibration does the reverse: hypothetical and risk-theoretical P&L move together, so attribution passes, while exceptions pile up because the level is wrong.

The combination is more powerful than either, and it is the closest the framework comes to actually validating a model. It is still weak in the sense Section 38.16 established (250 days, and Example 39.5's permissiveness), but the two tests fail on different errors, so a model that survives both is either wrong only moderately or wrong in what neither measures, such as the shape of the tail beyond the quantiles 250 days can test.

39.8 DESK-LEVEL APPROVAL, AND THE CLIFF

Approval is granted and withdrawn *desk by desk*, which is the framework's most important structural choice. Figure 39.4 sets out the sequence of tests a desk passes through.

A failing desk does not disqualify the bank, so the consequence is proportionate. But a red desk, or one that fails its backtest, moves to the standardized charge, which for a book with genuine diversification can run well above the model number. An amber desk keeps its model but pays a surcharge: half the excess of the standardized over the modeled charge for the green and amber desks together, scaled by the amber desks' share of the standardized charge. The total is in turn capped at what the standardized approach would require for every desk, except that any excess of the approved desks' modeled charge over their own standardized charge is still held, so the only saving the cap can deliver is the diversification the standardized calculation itself recognizes when all desks are computed together.

The move to red is a cliff, not a slope, and it is triggered by a statistical test on 250 days, which Section 38.16 already showed to be a short sample. So a desk's capital can rise by two-thirds (from 60 to 100 for the desk of Example 39.7 below) on evidence that would not settle a scientific question, and the rational response is to run risk models that are conservative in the specific sense of containing every factor the front office prices with, whether or not they matter. The amber surcharge softens the first step down without removing the cliff at the bottom.

That incentive is the framework working as intended. It cannot verify a model, so it makes the cost of divergence high enough that banks align their risk models with front-office pricing

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

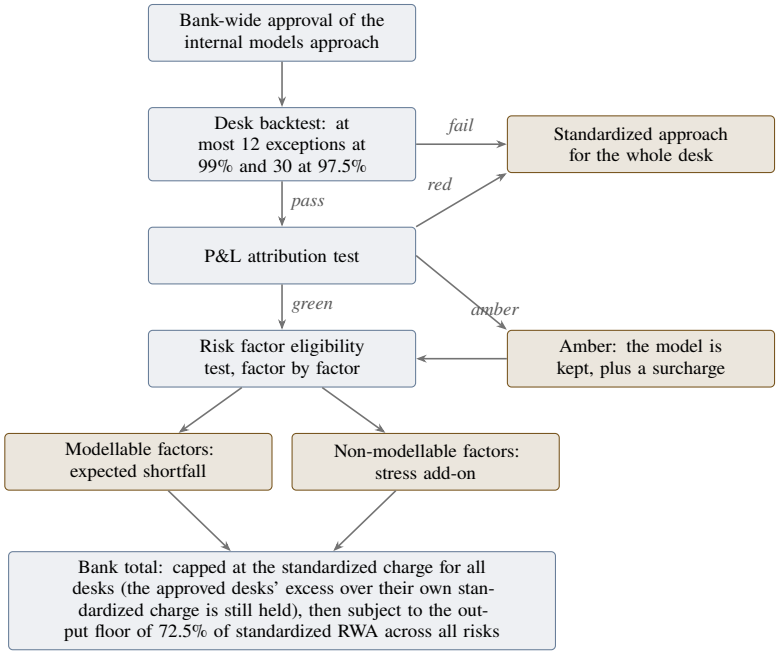


Figure 39.4: The tests a trading desk passes through, and where each outcome leads.

themselves.

39.9 ASSEMBLING THE TOTAL

The pieces above combine in a way worth seeing once, because the arithmetic explains where a trading book's capital actually comes from. The modeled part is

$$\max\Big(\mathrm{IMCC}_{t-1} + \mathrm{SES}_{t-1},\ \ m_c \times \overline{\mathrm{IMCC}} + \overline{\mathrm{SES}}\Big),$$

where IMCC is the internally modeled capital charge of Section 39.4, SES the non-modellable stress add-on, $t-1$ marks the previous day's figure, the bars denote sixty-day averages, and $m_c \geq 1.5$ is a multiplier that rises with the bank-wide backtesting exceptions of Section 39.7.

Example 39.6 Where the capital comes from

Take a bank with $\mathrm{IMCC}_{t-1} = 420$, $\overline{\mathrm{IMCC}} = 380$, $\mathrm{SES}_{t-1} = 95$ and $\overline{\mathrm{SES}} = 88$:

$$\max(420 + 95,\ \ 1.5 \times 380 + 88) = \max(515,\ 658) = 658.$$

The averaged, multiplied term binds, which is normal: with a multiplier of at least 1.5 it starts half again above a typical day's charge, so the requirement depends on a sixty-day average rather than on yesterday. Only a sharp rise in the latest figure reverses that.

The multiplier is where backtesting enters. The bank-wide exception count over 250 days sets m_c, with the zone boundaries of Section 38.16's traffic light, whose yellow zone the current framework calls amber:

Multiplier	Modeled charge	Increase
1.50 (green)	658	—
1.70 (amber, five exceptions)	734	+11.6%
2.00 (red)	848	+28.9%

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Adding the default risk charge on the approved desks and the standardized charge on the desks with no model approval gives the total. The residual risk add-on belongs to the standardized charge, so it sits inside the standardized-approach line:

Component	Amount	Share
Modeled measure (ES and NMRF)	658	60.2%
Default risk charge, approved desks	140	12.8%
Desks on the standardized approach	295	27.0%
of which residual risk add-on	35	3.2%
Total	**1,093**	100%

More than a quarter of the requirement comes from desks whose models were not approved, and that share is the framework's real lever. Hedging moves this number at the margin; passing the attribution test on more desks moves it wholesale—which is why the test of Section 39.6, and not the measure of Section 39.4, is what trading-book capital management is actually about.

39.10 THE OUTPUT FLOOR

One provision sits above everything in this chapter and can nullify all of it. The **output floor** caps the benefit of internal models across *all* risk types together: a bank's total RWA may not fall below 72.5% of what the standardized approaches would produce, once the floor is fully phased in: by 2028 under the Basel timetable, by 2030 in the European Union and the United Kingdom, and in China from 2024, where it replaced an earlier 80% floor. Canada has held its floor at 67.5% since 2025. It is not a market risk rule, which is exactly why it matters here: the trading book's modeled capital is one input to a constraint that binds at the level of the whole bank.

Example 39.7 The same desk failure, three banks, three costs

Each bank has standardized RWA of 100 for trading and 900 for everything else, so its floor is $0.725 \times 1{,}000 = 725$. Its trading book is a single desk, whose RWA are 60 on models and 100 on the standardized approach.

	Modeled RWA	Floor	Held
Bank A — modeled non-trading RWA 700			
Desk keeps approval	760	725	760
Desk loses approval	800	725	800
Cost of failing			**40**
Bank B — modeled non-trading RWA 600			
Desk keeps approval	660	725	**725**
Desk loses approval	700	725	**725**
Cost of failing			**0**
Bank C — modeled non-trading RWA 640			
Desk keeps approval	700	725	**725**
Desk loses approval	740	725	740
Cost of failing			**15**

The identical desk, failing the identical test, costs one bank 40, another 15 and the third nothing. For Bank B the floor already binds whatever the desk does, so the trading book's modeled number is slack and improving it changes no requirement at all. Bank C sits between: the floor absorbs 25 of the desk's 40, and only the part that lifts the bank above the floor is a cost.

This inverts the incentive the rest of the chapter carefully constructs. Sections 39.6 and 39.8 argued that the attribution test disciplines banks because failing it is expensive. For a floored

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

bank failing it costs nothing, and every other refinement earns nothing: a better model, a narrower NMRF population, a tighter liquidity classification. The tests still run, and they no longer change the requirement.

This explains a pattern that otherwise looks irrational. Section 39.13 notes that banks have chosen standardized capital over model approval on some desks. For a bank near the floor that is simply correct: model approval carries real operational cost and, below the floor, delivers no capital benefit whatever. For such a bank the framework's preferred approach is unattractive not because the modeling is hard but because a different rule has already decided the answer.

Whether this is a defect depends on what the floor is for. It exists because supervisors do not believe model outputs are comparable across banks: the same distrust that produced every test in this chapter, applied at the top rather than desk by desk. Having both is redundant by construction, and the redundancy is deliberate: the tests police the model, and the floor limits the damage if the tests fail.

39.10.1 Where the modeled credit number comes from

Example 39.7 took each bank's modeled non-trading RWA as given. Most of it is credit risk in the banking book, and for a bank approved to use the **internal ratings-based approach** (IRB) it comes from a formula that is Section 24.14's worst-case default rate with the supervisor's parameters in it. The bank supplies each exposure's probability of default (PD), its loss given default (LGD) and its effective maturity M; under the foundation version it supplies only the PD, and the LGD of a senior unsecured claim on a nonfinancial corporate is set at 40%. The capital per dollar of exposure is

$$K = \text{LGD}\left[N\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(0.999)}{\sqrt{1-\rho}}\right) - \text{PD}\right]\frac{1 + (M-2.5)\,b}{1 - 1.5\,b} \tag{39.1}$$

and the risk weight is $12.5K$, where 12.5 is the reciprocal of the 8% minimum capital ratio; ρ is a prescribed correlation and b a maturity coefficient, both given below.

Three choices in (39.1) are the supervisor's rather than the bank's. The confidence is 99.9% over one year. The correlation (the R of the Basel text, written ρ here as in Section 24.14) is prescribed rather than estimated, falling from 0.24 for the safest corporate borrowers to 0.12 for the riskiest,

$$\rho = 0.12\,w + 0.24\,(1-w), \qquad w = \frac{1 - e^{-50\,\text{PD}}}{1 - e^{-50}}$$

on the view that a strong firm defaults mostly when the economy does and a weak one mostly for reasons of its own. The last factor is a **maturity adjustment**, with $b = (0.11852 - 0.05478\ln\text{PD})^2$, which equals one at a one-year maturity and charges a longer loan for the downgrade risk that Example 24.10 measured. The PD itself is floored at 0.05%.

The formula can be applied one loan at a time because, in a one-factor model of an infinitely granular portfolio, the portfolio's quantile is the sum of the loans' own worst-case losses. That property is what lets a supervisor write a capital rule at all, and it is also the formula's blind spot: a book concentrated on a few large borrowers or exposed to a second systematic factor, a region or an industry, holds more risk than the sum says. Neither is charged in the formula; both are left to the supervisor's own review of each bank, outside the minimum requirement.

Example 39.8 Where the floor bites on a loan book

Take a corporate loan at each rating, with the one-year default probability from the last column of Table 24.2, an LGD of 40% and a maturity of 2.5 years, and compare its IRB risk weight with the standardized risk weight for a rated corporate in Table 27.1.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Rating	PD	ρ	WCDR	Maturity adj.	IRB RW	Std. RW	Ratio
A	0.05%	0.237	2.04%	1.752	17.5%	50%	0.35
BBB	0.28%	0.224	6.89%	1.411	46.7%	75%	0.62
BB	1.52%	0.176	16.95%	1.222	94.2%	100%	0.94
B	6.45%	0.125	32.42%	1.121	145.6%	150%	0.97
CCC	30.60%	0.120	72.59%	1.053	221.1%	150%	1.47

The BBB row runs as follows:

$$w = \frac{1 - e^{-50\times 0.0028}}{1 - e^{-50}} = 0.1306, \qquad \rho = 0.12 \times 0.1306 + 0.24 \times 0.8694 = 0.2243,$$

$$\text{WCDR} = N\left(\frac{N^{-1}(0.0028) + \sqrt{0.2243}\, N^{-1}(0.999)}{\sqrt{1 - 0.2243}}\right) = N(-1.4837) = 6.895\%,$$

$$b = (0.11852 - 0.05478 \ln 0.0028)^2 = 0.1941,$$

$$K = 0.40 \times (0.06895 - 0.0028) \times \frac{1 + (2.5 - 2.5) \times 0.1941}{1 - 1.5 \times 0.1941} = 3.733\%,$$

and the risk weight is $12.5K = 46.7\%$.

The ratio column is the one that matters for the floor. A loan rated A or BBB is modeled at well under 72.5% of its standardized weight (35% of it at A, 62% at BBB), so a bank whose book is mostly A and BBB lending, the typical book of a large bank lending to large companies, sits on the floor whatever its models say. Speculative-grade loans are modeled at close to or above their standardized weight, well above 72.5% of it, and leave the floor slack. Bank B of Example 39.7 is therefore not a curiosity. It is what a bank with a high-quality loan book looks like, and for such a bank the trading desk's model approval is worth nothing because its lending has already put it on the floor.

The standardized side matters too. The United States does not use credit ratings in its current capital rules, and its standardized approach puts most corporate exposures at 100%, against which even the BBB loan is modeled at under half, so the floor bites harder there. The European Union went the other way and, until 2032, lets a bank's floor calculation weight an unrated corporate exposure at 65% where its own PD estimate is no more than 0.5%, precisely to soften the floor on investment-grade lending.

Figure 39.5 draws the whole curve. The IRB curve runs below 72.5% of the standardized weight at both the A and BBB points of the Basel table, so that is where a loan book pushes a bank onto the floor.

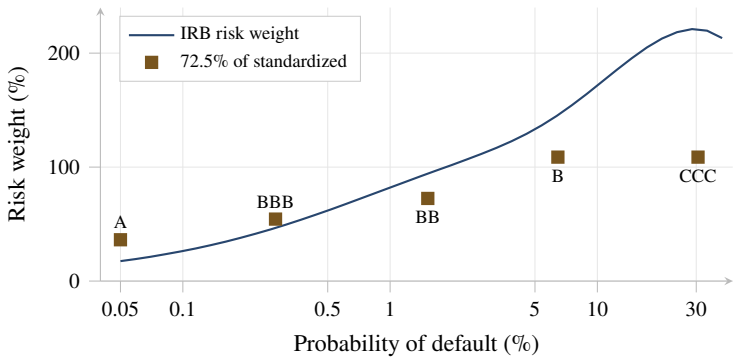


Figure 39.5: IRB risk weight for a corporate loan (LGD 40%, maturity 2.5 years) against 72.5% of the standardized weight at each rating's default probability.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

39.11 PRUDENT VALUATION

There is a second capital channel that has nothing to do with the measures above, and for an exotic book it can matter more. In the European Union and the United Kingdom it is a rule; elsewhere supervisors expect valuation reserves of the same kind without prescribing them.

Fair value is an exit price, typically marked at mid within the bid–ask spread or at a model price. Prudent value is a price the position could actually be realized at with 90% confidence, and the difference, the **additional valuation adjustment** (AVA), is deducted from common equity directly rather than risk-weighted.

Example 39.9 The deduction on a $100 million book

An exotic book with a fair value of $100 million carries the following adjustments. Three of them, market price uncertainty, close-out cost and model risk, overlap in what they measure, so the core approach of the European rules, as against the simplified approach for small books, adds them and then counts half of the sum. The rules split the unearned-credit-spread and funding adjustments into the same three parts, so they join that sum.

Component	Basis points	Amount
Market price uncertainty	40.0	$400,000
Close-out cost (bid–ask)	35.0	$350,000
Model risk	60.0	$600,000
Unearned credit spread	10.0	$100,000
Investing and funding costs	8.0	$80,000
Sum, counted at 50%	76.5	$765,000
Concentrated position	25.0	$250,000
Future administrative costs	6.0	$60,000
Early termination	5.0	$50,000
Operational risk	4.0	$40,000
Total	**116.5**	**$1,165,000**

Model risk is the largest single component in this book, and it is the one a desk cannot reduce by trading better: a tighter market narrows the close-out cost and does nothing to the model uncertainty of a payoff nothing quotes.

The capital arithmetic makes the size clear. A $1.165m deduction from common equity, at an assumed total requirement of 12% of RWA, removes the capital that would support $9.7m of risk-weighted assets, equity the bank can no longer use for anything else, whatever the position's own risk weight.

This is where Section 21.11's model spread becomes a number on a balance sheet. Two models agreeing on every vanilla and disagreeing on an exotic produce a valuation uncertainty; that uncertainty is not a modeling curiosity but the input to a deduction, and it is why dealers maintain model reserves against exotic books instead of relying on a single model's mark.

The measurement that makes this possible comes from Hull and Suo (2002), and the regulatory framing arrived afterward. Their procedure (calibrate competing models to a common surface, verify each reproduces it, then price what the surface does not determine) turns model risk from a caveat into a quantity, and a quantity is the only thing a capital rule can act on. The prudent valuation regime is that idea written as a requirement: it does not ask whether a model is right, which no rule could adjudicate, but how far defensible alternatives disagree.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

39.12 STRESS TESTING AND WHAT SITS OUTSIDE THE FORMULA

Everything above is a formula, and supervisors do not rely on formulas alone.

Supervisory stress tests apply a common scenario across banks and compare. Their value is the comparison rather than the scenario: a bank whose losses are unusual under a shared shock is revealing a concentrated position, and the exercise finds it without anyone specifying what to look for. In the United States the exercise also sets capital: the largest banks' trading books take an instantaneous global market shock, and the losses feed the **stress capital buffer**, a requirement on top of the minimum that rises with the bank's projected stress losses. For those banks a supervisory scenario, not the formula above, often decides how much the trading book costs.

Reverse stress testing inverts the question—not "what do we lose in this scenario" but "what scenario makes us fail"—and it is the only exercise in the framework that begins from the outcome. It is also the one most likely to find Section 38.19's blind spots, because it is not constrained to the risk factors the model contains. Section 38.17 works one through for a two-asset book, and finds that the most plausible route to a given loss runs through the position carrying the largest share of the VaR.

And the supervisory review sets capital above the formula where the formula is known to be inadequate; concentration, which Section 38.19 said no distribution-based measure sees, is the standard case.

The pattern across all three is the same. The formula handles what can be specified in advance; the discretionary machinery handles the rest, and the rest is where Chapter 40's losses live.

39.13 WHAT THE FRAMEWORK CHANGED, AND WHO BEARS THE COST

Three things changed, and only the first was intended. Capital went up for trading, and unevenly. The increase falls hardest on exotic and illiquid books (through the residual risk add-on, the NMRF treatment, and the long liquidity horizons) and lightly on high-volume trading in liquid instruments.

The internal models approach became less attractive than it was designed to be. Between the operational cost of the attribution test, the NMRF data requirements, and the cliff risk, many banks concluded that standardized capital on some desks was cheaper than model approval (the PRA reported in June 2026 that few UK firms planned to adopt the internal models approach), which is the opposite of the intended direction, and Section 39.10 showed why the floor makes that conclusion correct for some of them. A framework that makes its own preferred approach unattractive has a design problem.

The business consolidated. Fixed costs favor scale: the data infrastructure for modellability, the daily attribution machinery, and the desk-level governance are nearly independent of volume. Chapter 35's structured issuance and Chapter 27's client clearing both concentrated for this reason, and the trading book is a third instance. The cost lands in four places.

Dealers bear it directly and pass what they can into spreads, which is why the pricing of Chapter 35's structured issuance and Chapter 34's exotics carries a capital component that the models in those chapters do not contain.

Clients bear it as a wider bid–ask on precisely the trades the framework taxes (bespoke, illiquid, long-dated), and the effect is to push demand toward standardized products, which is intended.

Smaller dealers bear it as a fixed cost and some exit, so the framework's second-order effect

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

is concentration, and Section 27.12's concern about concentration is partly a consequence of the rules in this chapter.

And supervisors bear the residual: a framework that cannot verify models must set incentives instead, so the quality of the outcome depends on the calibration of penalties rather than on the correctness of any measurement.

39.14 MODEL RISK

Everything above assumes the model maps market factors to portfolio value correctly, and that assumption is itself an exposure. *Model risk is not a subset of market risk*, which is why it needs its own treatment. A market risk measure is conditional on the pricing model: the model that marks the position also supplies the revaluations its risk is measured from, and the attribution test compares the risk model with that same pricer, never the pricer with anything outside it. An error in the pricer misstates the mark and the risk at once, and nothing inside the system disagrees with it.

Section 21.11 made this concrete. Two models calibrated to identical vanilla prices give different values for the same exotic, and the difference is not measurement error—it is the part of the payoff the calibration instruments do not span. No amount of data resolves it, because the disagreement is about what happens where nothing trades.

The governance response is well established and modest in its claims. Models are inventoried; each has an owner and a stated purpose; validation is performed independently of development; limitations are documented and the documented limitation is the basis for a reserve or a capital add-on. In the United States the supervisory guidance is SR 26-2 (2026), which replaced the 2011 guidance known as SR 11-7 and scales the effort to the risk a model creates; in the United Kingdom it is the PRA's SS1/23, in force since May 2024. Neither document verifies a model, and neither claims to: they make the assumptions explicit and assign someone to own them.

39.15 VALIDATING WHAT CANNOT BE VERIFIED †

A model is *verified* when its output can be checked against something independent: an analytic formula, a converged numerical method, a market price. Validation is the weaker activity of testing a model that cannot be checked that way, and two kinds of model force the distinction.

The first is the calibrated model of Section 21.11. Local and stochastic volatility models fitted to the same vanilla surface agree on every vanilla and disagree on a barrier, and nothing that trades can say which is right. Validation can only measure the disagreement, which is what Hull and Suo's procedure and Section 39.11's model risk adjustment do, and cannot remove it.

The second is the learned model. A **learned pricer** is a neural network trained to reproduce a pricing model's outputs, so that a slow calculation can be evaluated in microseconds; a **learned hedging policy** is a network trained on simulated price paths to choose positions that minimize a risk measure such as Chapter 38's expected shortfall. The online note [TN41] treats both, and three of its findings bear on validation.

The training set is part of the model. A learned pricer degrades silently outside the range of inputs it was trained on, returning a plausible number rather than an error ([TN41], Section 3). A model inventory that records the network and omits the training range has recorded the less important half, and retraining on new data is a model change requiring revalidation, not a refresh.

Reproducibility is not the same as determinism. In Section 12 of [TN41], three training runs of one hedging policy differed only in the random number seed, agreed on the expected shortfall they achieved to within 0.8%, and held positions that differed by up to 1.7 percentage

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

points of delta. A validation standard that asks whether results reproduce will pass all three, although the positions they would actually hold differ.

Validation reduces to out-of-sample testing in the region someone happened to sample. A Black–Scholes implementation can be checked against an analytic formula; a learned pricer can only be checked against the pricer it learned from, and the case study in Section 4 of [TN41] is the case where that pricer was wrong and the network smoothed the evidence.

The existing framework is adequate in structure and strained in practice. Inventory, ownership, independent validation, and documented limitations are the right categories, and SR 26-2 applies them to non-generative machine-learning models such as these, leaving only generative and agentic artificial intelligence (AI) outside its scope. What is missing is a validation technique with the power to distinguish a good model of either kind from a plausible one, and until there is one, the sensible response is the conservative one: use such models where a classical answer exists to check them against.

SUMMARY

A capital rule is a contract about a number the bank itself produces, so the framework both specifies the model and tests it. Each complication answers a specific failure: the prescriptive boundary removes an arbitrage, stress calibration answers procyclicality, and liquidity horizons make the horizon depend on what is held, though only by risk-factor class. A desk whose horizons are spread across the schedule pays materially more than under a flat ten-day measure with no change in any position.

The standardized charge refuses to trust a correlation. It takes the worst of the prescribed correlations, a higher set and a lower set, and which scenario binds is a property of the book: a directional position is worst under high correlation and a butterfly under low, by a far wider margin. Curvature follows Chapter 18's economics: long optionality attracts no charge, and the same option sold attracts all of it. Non-modellable factors are capitalized by a stress scenario added without diversification, so a book with a sizable non-modellable share pays markedly more capital for identical risk.

The P&L attribution test is more permissive than its reputation. A desk may leave a sizable share of its P&L variance unexplained and stay green, because both statistics leave the green zone only once a missing factor carries roughly a third of the variance, so the test catches a large missing factor but not a moderate one. Desk-level backtesting checks the model's calibration and attribution its content, and a model that passes both is wrong only moderately or in what neither sees. Approval is withdrawn desk by desk against a 250-day sample, so capital moves as a cliff, softened only by the amber surcharge.

The output floor holds risk-weighted assets at no less than 72.5% of the standardized result when fully phased in, so for a bank where the floor already binds, failing the attribution test costs nothing, which is why declining model approval can be correct. Most of the modeled number the floor compares is banking-book credit, set by the IRB formula, which is the worst-case default rate at 99.9% with a prescribed correlation and a maturity adjustment; it models A and BBB lending at well under 72.5% of the standardized weight, so the floor binds hardest on banks with the best loan books. Prudent valuation deducts from equity directly, and for an exotic book model risk can be its largest component. Supervisory stress tests and review set capital where the formula cannot, and in the United States a stress scenario often decides the trading book's cost.

Model risk is not a subset of market risk: the pricing model that marks a position also feeds its risk measure, so an error in it misstates both at once, and no test in the framework compares it with anything outside itself. Governance makes assumptions explicit without verifying anything, and models that cannot be verified (calibrated models that disagree where

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

nothing trades, and learned models) strain it most, so the conservative use is the defensible one.

FURTHER READING

The Basel Committee's *An Explanatory Note on the Basel II IRB Risk Weight Functions* (2005) derives the formula of Section 39.10 and explains each of its parameter choices, and Gordy (2003) proves the portfolio invariance that lets it be applied one loan at a time. The Basel Committee's *Minimum Capital Requirements for Market Risk* (2019) is the primary source for everything from Section 39.2 to Section 39.9; the sections motivating the boundary and the liquidity horizons are the ones to read first, and the industry quantitative impact studies are where the capital effects of Section 39.13 are actually measured. The output floor is in the Committee's *Basel III: Finalising Post-Crisis Reforms* (2017), which is the document to read for how the floor was meant to interact with internal models. On prudent valuation, the European Banking Authority's regulatory technical standards, adopted as Commission Delegated Regulation (EU) 2016/101, set out the nine adjustments and their aggregation. Jurisdictional implementation differs in ways that matter and in ways that do not; the divergences worth tracking are the treatment of internal models and the boundary, and both are documented in supervisory consultation papers rather than in the standard. On model risk, the US interagency guidance SR 26-2 (2026) and the PRA's SS1/23 (2023) are the current frameworks, and the earlier SR 11-7 (2011) is worth reading for the vocabulary both inherited and because it makes no claim to verify anything. Hull and Suo (2002) is the paper behind Section 39.11's measurement of model risk. On the questions raised in Section 39.15, SR 26-2's footnote on generative and agentic AI models is where US supervisors currently draw the line.

PRACTICE QUESTIONS

39.1 Explain why the trading-book boundary was made prescriptive, compute the capital at stake in Section 39.2's bond portfolio, and explain why removing the capital benefit of reassignment matters more than tightening the definition.

39.2 Explain why the standardized approach supplies correlations rather than estimating them, and relate this to Section 38.11.

39.3 Using the weighted sensitivities and correlations of Example 39.1, compute the directional book's bucket charge under the high-correlation scenario, and explain why it equals the sum of the weighted sensitivities.

39.4 Using Example 39.1, explain why the directional book is worst under high correlation and the butterfly under low, why the butterfly's charge falls to zero under high correlation, and why the hedged book nonetheless pays the larger proportional penalty.

39.5 Explain why a long option attracts no curvature charge while the same option sold attracts the whole of it, using Figure 39.2.

39.6 Compute the curvature charge on a short \$10 million at-the-money six-month put on the stock of Section 39.3 at 25% volatility and zero interest rates, and explain why it equals the short call's.

39.7 Recompute Example 39.2 for a small-cap stock, whose volatility has a 60-day liquidity horizon, keeping the curvature shock at 55% so that only the vega term changes, and state the total charge as a share of notional and the curvature term's share of it.

39.8 A desk holds \$50 million notional of variance swaps and \$200 million of products carrying gap risk. Compute its residual risk add-on, and explain why the framework charges a flat percentage rather than modeling these payoffs.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

39.9 In Example 39.3, a factor class is reclassified from a 40-day to a 60-day horizon, raising ES_4 from 45 to 60. Compute the new aggregate, and explain why the desk's capital changed although nothing it holds did.

39.10 State the risk factor eligibility test, and redo Example 39.4 with a correlation of 0.50 between the two parts, separating the cost of lost diversification from the cost of the stress calibration.

39.11 Using the closed form of Example 39.5, compute the share of unexplained variance at which the rank correlation equals 0.75, and state the zone the desk is in.

39.12 Explain why the rank correlation and the Kolmogorov–Smirnov statistic leave the green zone at nearly the same share of unexplained variance, and construct a model error that only one of them would detect.

39.13 Explain what kind of model failure the attribution test catches and what kind it does not, and relate the second to a learned pricing or hedging model.

39.14 A desk records 9 exceptions at 99% and 34 at 97.5% over 250 days. State the consequence, and explain why VaR is backtested when capital is held against expected shortfall.

39.15 Explain what backtesting tests that attribution does not, and construct a desk that passes each while failing the other.

39.16 Explain why desk-level approval is a cliff rather than a slope, what the amber surcharge changes, and what behavior the cliff induces.

39.17 Using Example 39.6, recompute the modeled charge when yesterday's internally modeled charge is 650, state which term binds, and explain why the averaged term usually does.

39.18 State how the bank-wide backtesting exception count enters the capital formula, compute the increase from a green multiplier to the amber multiplier for seven exceptions (1.83), and explain why the increase is smaller than the rise in the multiplier.

39.19 Using Example 39.6, state the share of the total requirement contributed by desks without model approval, and explain why this makes the attribution test the focus of capital management.

39.20 Using Example 39.7, find the modeled non-trading RWA at which failing the desk's test costs exactly 20, and explain what the floor does to the incentives of Section 39.6.

39.21 Explain why a bank near the output floor may rationally decline model approval, and why this is not evidence that its modeling is inadequate.

39.22 Using Example 39.9, compute the deduction if the adjustments in the 50% sum were added in full rather than at 50%, explain why model risk cannot be reduced by trading better, and compute the risk-weighted assets the full deduction would displace at a 12% capital requirement.

39.23 Explain how a supervisory stress test becomes trading-book capital in the United States, and why reverse stress testing is the exercise most likely to find what the formula misses.

39.24 Explain why model risk is not a subset of market risk, and why the error and the measure move together.

39.25 Using Section 39.15, explain why reproducibility is an inadequate validation standard for a learned hedging policy, and what should be required instead.

39.26 A senior unsecured loan to a nonfinancial corporate has a one-year PD of 1% and a maturity of 2.5 years, and the bank uses the foundation IRB approach. Compute the prescribed correlation, the worst-case default rate, the maturity adjustment, the capital

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

requirement and the risk weight, and state whether the output floor binds on a book of such loans if the standardized weight is 100%.

39.27 Using Example 39.8, explain why the output floor binds on a bank that lends mainly to investment-grade companies and not on one that lends mainly to speculative-grade companies, and what this implies for the value of trading-book model approval at each.

FURTHER QUESTIONS

39.28 Example 39.5 assumes the unexplained component is independent noise. Redo the analysis with an unexplained component that is systematically related to the explained one (a missing factor correlated with the ones present) and determine whether the test becomes more or less powerful.

39.29 Example 39.5 shows the two statistics reacting to a missing factor at nearly the same point. Construct a desk whose model error makes the Kolmogorov–Smirnov statistic bind well before the rank correlation, and state whether that error is more or less serious than the one the correlation catches.

39.30 Section 39.8 argues the cliff induces banks to include factors that do not matter. Model the bank's optimization explicitly (capital cost against the probability of failing the test) and determine the conditions under which adding an immaterial factor is rational.

39.31 Example 39.3 treats liquidity horizons as a fixed schedule. Estimate empirical close-out horizons for three instrument classes, compare them with the regulatory assignment, and determine the capital effect of the difference.

39.32 Section 39.13 argues that fixed costs concentrate the market. Build a model of dealer entry and exit under a fixed compliance cost, calibrate it to the observed consolidation, and determine what level of fixed cost the observed market structure implies.

39.33 Section 39.15 argues that no current validation technique separates a good learned model from a plausible one. Propose one, state what it would need to observe, and determine whether the required data exists for a pricing surrogate and for a hedging policy respectively.

39.34 The framework taxes illiquidity because it cannot model it. Design a capital treatment that charges for liquidity directly, state what it would require banks to report, and determine whether the reporting burden is smaller or larger than the current non-modellable-factor machinery.

39.35 Section 39.10 shows a floor-bound bank indifferent to its desk tests. Model a bank's choice, desk by desk, between internal models and the standardized approach as a function of its distance from the floor, and determine the distance at which the internal models approach stops paying for its fixed cost.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Chapter 40
Derivatives Mishaps and What We Can Learn

This chapter is not a collection of cautionary tales. Every case in it is a natural experiment, one the profession never meant to run, and each one tested a specific assumption made somewhere earlier in this book.

One finding organizes the chapter and it is not the expected one. In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed. What failed was around them—the cash, the exit, the information, the venue, the buyer's ability to price what it bought, or the decision not to hedge at all. The last of those deserves saying at the start: the most recent case, Section 40.14, is a bank that failed because it did not hedge, and any chapter that presents derivatives as the source of danger has the sign wrong.

Chapters 38 and 39 built the measures and the rules; most of these losses fell in the gaps Section 38.19 lists, and Chapter 39's controls are read here against the cases behind them. Sections 40.1 to 40.14 take the cases one at a time; Section 40.15 draws out what repeats, Section 40.16 what changed, and Section 40.17 what institutions and regulators should take away.

40.1 METALLGESELLSCHAFT: CORRECT IN VALUE, WRONG IN TIME

In 1993 a German industrial group's US subsidiary had sold long-dated fixed-price oil supply contracts and hedged them with a stack of near-dated futures and short-dated swaps, rolled forward each month. The hedge was sound in the only sense a valuation model recognizes. Whether the parent was right to liquidate it rather than fund it is still argued, and Chapter 3's case study names both sides of that debate; what neither side disputes is that the cash the hedge demanded in a fall had not been committed in advance.

Example 40.1 The cash flow and the value point in opposite directions

Take the historical position, 160 million barrels of forward commitment, and treat it as hedged one-for-one with front-month futures; the real stack mixed exchange-traded futures with short-dated over-the-counter swaps. The price falls and the roll spread below are illustrative:

Price fall	Futures margin (cash out)	Supply contract (unrealized)
\$2.00/bbl	−\$0.32bn	+\$0.32bn
\$5.00/bbl	−\$0.80bn	+\$0.80bn
\$8.00/bbl	−\$1.28bn	+\$1.28bn

Every row is flat in undiscounted value and catastrophic in cash. The futures loss is paid in variation margin now; the supply contract's offsetting gain arrives over ten years and cannot be pledged. Chapter 3's case study records the historical outcome, margin calls of roughly \$0.9 billion by the end of 1993 and losses of about \$1.3 billion.

The roll was a second exposure no one had chosen. A stack-and-roll pays or earns the calendar spread every month on the whole remaining stack, which converts a flat-price hedge into a calendar spread position; Section 36.10 works that arithmetic. On this position, with deliveries running the stack down to an average of about 153 million barrels over the first year, a contango of $0.35 a barrel per roll costs about $0.64 billion that year while the flat-price hedge worked perfectly.

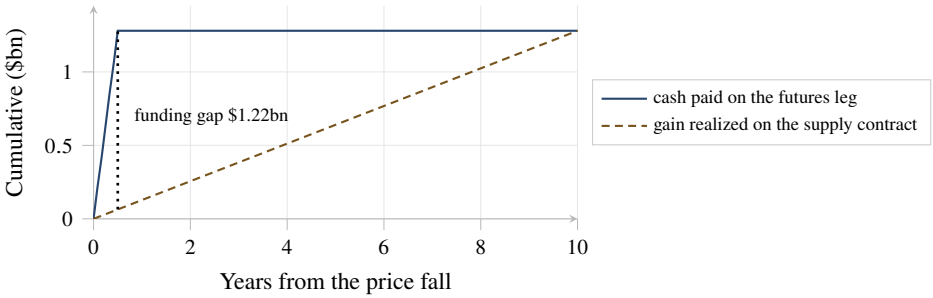


Figure 40.1: An $8/bbl fall spread over six months, on Example 40.1's position; illustrative.

Figure 40.1 is the mechanism of this chapter in one picture. Both curves reach $1.28bn, so the hedge is exact in undiscounted terms—and the vertical distance between them is a loan the strategy never arranged. The gap peaks at $1.22 billion at six months, once the margin has all been paid, and it closes only at the ten-year horizon.

Every case that Section 40.15 groups under cash and value is that gap. The LDI schemes, the 2022 European generators, the margin calls at AIG, and Example 37.2's basis trade all draw the same two curves with different labels. A valuation model computes where the curves end; the failure happens in between.

The lesson is Section 8.9's: margin calls are a liquidity risk, and Section 8.10's stress test measures them because it counts gross cash outflow, not net loss. On Example 40.1's position it reports $0.16 billion of variation margin for every dollar the oil price falls, where a value measure reports nothing. The funding line should be committed at that scale before the trade, and the roll budgeted as a calendar spread.

40.2 PORTFOLIO INSURANCE, 1987: WHEN THE HEDGE IS THE FLOW

This is the clearest case in the chapter of a hedging technique moving the market it was hedging against (the LDI selling of Section 40.8 is another). Section 16.2 introduced the technique; this is the episode and its arithmetic.

Portfolio insurance replicated a protective put synthetically: rather than buy a put, sell index futures as the market falls and buy them back as it rises, in the proportion Chapter 18's delta prescribes (Sections 18.11–18.12). The replication is sound in theory, and the programs followed it as closely as the market allowed (on October 19, 1987 they fell behind schedule because the futures market could not absorb the selling), so the mechanism failed, not the execution.

Example 40.2 The selling schedule was published by the mathematics

Take $70 billion insured with a floor 5% below spot over three months at 20% volatility, with the interest rate set to zero for simplicity. The programs must be short $-\Delta$ of the index on a portfolio whose own value falls with the index, so the futures short is $|\Delta| \times \$70\text{bn} \times S/100$ at index level S:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Index	Put delta	Futures short	Futures sold
100	−0.287	\$20.07bn	—
95	−0.480	\$31.92bn	+\$13.18bn
90	−0.688	\$43.35bn	+\$13.48bn
80	−0.952	\$53.33bn	+\$15.89bn

A 10% decline obliges the programs to sell \$26.7 billion, and a 20% decline \$42.5 billion, with no discretion at any point. That is more than the change in the short, because the contracts already held lose dollar value with the index, and that loss needs no trade.

The selling per point of the index is driven by gamma, so it is fastest near the floor and slows once the delta approaches −1; the last row covers ten points and the two above it five each. Near the floor the delta changes by about 0.042 per index point, which, on the 0.70 billion index units insured at an index of 95, is about \$2.79 billion of futures per point. Figure 40.2 draws the whole schedule.

The critical feature is what a synthetic put changes. A purchased put has a *seller*. Left unhedged, that seller's position calls for no trading at all; a dealer who delta-hedges the short put is short gamma and *sells* into the decline as well, as Section 18.12 shows. Listed options therefore spread the selling across writers, not all of whom hedge; they do not abolish it. Replication puts the whole of it in the holders' own programs, on a schedule anyone can compute in advance. The protection was real for each individual holder and, in aggregate, was manufactured out of the very selling that deepened the decline it insured against.

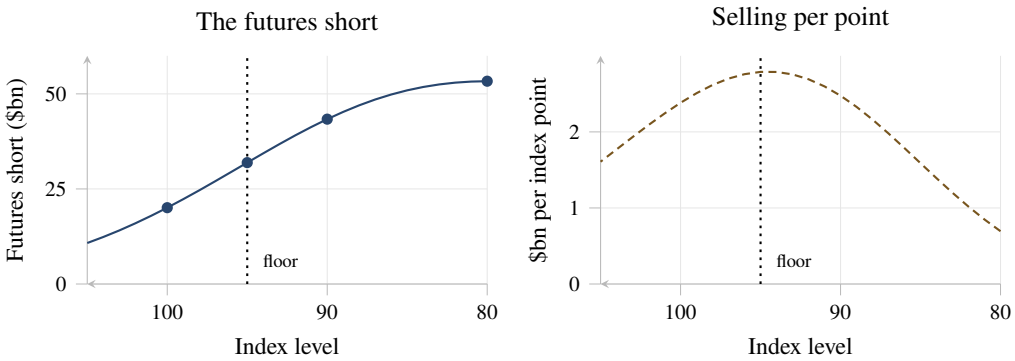


Figure 40.2: The portfolio insurers' selling schedule in Example 40.2, as the index falls from 105 to 80.

The mechanism has appeared three more times in this book. Section 35.6's autocallable issuers buy into a decline and then, at the barrier, sell about a fifth of the hedge at once; Section 37.3's liquidation cascade forces sales as a published function of price; and Section 23.7's dealer gamma positioning is the same arithmetic run daily on a much shorter horizon. In each, hedging demand is a mechanical function of the price, so at the level that matters the hedge and the move reinforce each other. That is the procyclicality of Section 38.18 with a hedge ratio in place of a risk measure: there a capital requirement that rises after volatility arrives forces positions to be cut into a falling market, and here a delta that grows as the index falls does the same. In both, the feedback needs the rule to be shared. One program selling on its own delta moves nothing; \$70 billion of insured portfolios following the same rule at once moved the index, and the move set the next round of selling.

What changed afterward is worth noting because it was not a ban. Circuit breakers were introduced (a mechanism that interrupts the feedback rather than the strategy), and the listed index option market grew. The fix was to make the protection purchasable rather than

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

manufacturable, and that is the correct response to a mechanism rather than to an instrument.

Section 18.12's gamma arithmetic publishes a replicating hedge's flow in advance: the futures sold per index point (gamma times the insured portfolio's value) peaks in Example 40.2 at about $2.79 billion at the floor (Figure 40.2). A program should set that number, summed over everyone running the same rule, against what the futures market can absorb, and where the market cannot absorb it, buy a listed put instead.

40.3 BARINGS AND SOCIÉTÉ GÉNÉRALE: A FAILURE OF CONTROLS, NOT OF DERIVATIVES

Two cases, thirteen years apart, are failures of the risk control system rather than of the instruments, and neither has any derivatives pricing content. In both, the checks that should have caught an unauthorized position (the **segregation of duties** that keeps trading apart from its settlement, independent reconciliation, and verification of what the trader reported) were controlled or defeated by the person they were meant to check.

In both, positions reported as low-risk arbitrage were outright directional. At Barings one person controlled trading and the settlement of that trading, and performed the reconciliation that would have exposed the position. At Société Générale the trader had come from the middle office and knew its checks well enough to hide the positions behind fictitious offsetting trades.

	Position	Firm equity	Ratio
Barings, 1995	$27bn	$0.6bn	**45**×
Société Générale, 2008	EUR 50bn	EUR 30bn	1.7×

The same failure destroyed one firm and did not destroy the other, and what decided the outcome was the ratio. Barings held about forty-five times its equity in directional futures and option positions and lost £827 million on them; the French bank held a larger notional against a balance sheet that could absorb the EUR 4.9 billion it cost.

The instruments were exchange-traded: index futures in both, and at Barings also Nikkei options and Japanese government bond and Euroyen futures, all cleared, margined daily, and priced as in Chapters 5 and 15. No model failed and no instrument surprised anyone.

Separating trading, settlement, and reconciliation is necessary and will still sometimes fail, and when it does, the severity is set by a number the controls do not watch: the largest possible position against equity. Section 2.3's daily settlement lets a firm cap that number, because every exchange-traded loss is paid in margin the next morning, and capping the margin funding a desk can draw caps the position. At Barings that funding was the visible signal: the head office kept remitting margin to Singapore for a position it believed was hedged.

40.4 LTCM: LEVERAGE PRICES THE PATH, NOT THE VIEW

The 1998 failure of Long-Term Capital Management (LTCM) is usually told as a story about overconfidence in models: a fund whose partners included two Nobel laureates trusted spreads and correlations estimated in calm markets, and the Russian default of August 1998 broke them. That story is not wrong, but the arithmetic tells a narrower and more useful one.

Example 40.3 A correct view at the wrong leverage

Equity of $4.7bn against a $125bn balance sheet is **26.6**× leverage, before derivatives notionals far larger than either.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Adverse move on assets	Loss	Share of equity
0.5%	\$0.63bn	13%
1.0%	\$1.25bn	27%
2.0%	\$2.50bn	53%
4.0%	\$5.00bn	**106%**

The wipeout move is \$4.7bn/\$125bn = 3.76%. So the strategy could fail with the convergence trades (long the cheaper of two closely related securities, short the richer, waiting for the gap to close) right: it needed them to converge without the balance sheet first losing 3.8% of its value. And convergence spreads often widen before they converge; bearing that widening is what the trade is paid for. At that leverage the fund was not expressing a view about value; it was expressing a view about the path, that spreads would not widen that far before converging, and it had not priced that path.

Leverage converts the path into a survival condition that a VaR at a horizon does not see, so size the trade by the path: Section 38.17's reverse stress test finds the adverse move the position must survive on the way to converging, 3.76% of the balance sheet in Example 40.3, and leverage should keep that move well beyond the widening such trades have shown. A position others also hold makes the move larger: the fund's forced sales mark down the others' books and they sell too, which is Section 40.2's shared rule in another form. LTCM's trades had been copied across the dealer community and were unwound together in the summer of 1998; on September 23 the New York Fed brokered a \$3.6 billion recapitalization by fourteen firms.

40.5 AMARANTH: EXIT COST IS A FUNCTION OF YOUR OWN SIZE

The hedge fund Amaranth's natural gas loss of about \$6.6 billion in September 2006 shows that concentration, which Section 38.19 says a distribution-based measure cannot see, is not an academic gap. The fund held positions approaching 40% of open interest in NYMEX's winter-month natural gas contracts. At that share, the exit is not a market price—it is a schedule. With daily volume taken, for illustration, as equal to the open interest of those contracts, the days to exit are 40% divided by the participation rate:

Maximum participation in daily volume	Days to exit
10%	4.0 days
20%	2.0 days
30%	1.3 days

Every one of those days is a day the market knows a forced seller is present. VaR is read from the distribution of price changes, which does not depend on how much of the market the holder owns; only the exit does.

Set the horizon of the measure by the holder's own exit, not by the risk factor's class. At a 10% participation rate the table's exit takes four days, so Section 38.14's square-root-of-time rule scales the one-day VaR to the four-day exit by a factor of two if the position is held whole until sold—and it still assumes prices unmoved by the seller, which at 40% of open interest they are not. A risk manager should therefore limit the share of open interest directly and stress the exit with an impact cost, since Section 39.4's liquidity horizons, assigned by class, will not.

40.6 ARCHEGOS: THE SAME RISK AT MANY BROKERS

In March 2021 Archegos Capital Management, a family office holding concentrated stock positions through swaps, collapsed: its capital was wiped out, and the dealers lost more than

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

$10 billion. It was a failure of measurement, not of modeling.

Example 40.4 Adequate margin against the wrong exposure

Take a family office holding concentrated single-stock exposure through total return swaps at six prime brokers. Each broker saw its own slice and margined it at a level it judged adequate for that slice. Section 27.14 works the per-broker arithmetic, which leaves $7.5 billion of margin across the system against a $50 billion gross position.

But the liquidation was not six independent liquidations. All six sold the same stocks within days, and since a broker could look only to the margin it held, only the margin stood against the loss:

Impact of simultaneous exit	Loss	Against $7.5bn margin
10%	$5.0bn	covered
25%	$12.5bn	**short by $5.0bn**
40%	$20.0bn	**short by $12.5bn**

The margin was set for the exposure each broker measured and was wrong for the exposure that existed. No broker's own book showed the difference, and Section 27.14 adds that the margin levels themselves were set by competition for the client's business rather than by the concentration of the positions.

The instrument choice was not incidental. A total return swap gives economic exposure without an ownership disclosure, so the aggregate was invisible by construction: the same property that made the product attractive made the risk unmeasurable. Section 27.13 argued that bilateral measurement has a limit, and this is the limit.

Margin a total return swap (Section 25.9) on the client's whole position and its exit, not on one broker's slice: Section 27.15's initial margin is meant to cover the close-out loss, and in Example 40.4 a joint exit at 25% impact loses $12.5 billion against $7.5 billion collected. A prime broker should ask for the positions held elsewhere, treat a refusal to disclose them as a risk in itself, and set the rate by Section 40.5's exit arithmetic, not by competition.

40.7 LME NICKEL: WHEN THE VENUE IS THE RISK

In March 2022 the nickel price, which had closed near $29,000 a metric ton on Friday, March 4, more than tripled in the next two trading days to above $100,000, against a very large short position held by a producer as a hedge. Take an illustrative 100,000-metric-ton short; the producer's actual position was larger, and much of it was held over the counter, out of the exchange's view.

Price	Loss on a 100,000-metric-ton short from $29,000
$40,000	$1.10bn
$50,000	$2.10bn
$80,000	$5.10bn
$100,000	$7.10bn

The exchange then canceled several hours of trades. That cancellation is the case's real content, and it tests the assumption Section 2.10 flagged as a judgment about institutional resilience rather than a fact. Every calculation in Parts I and V treats the clearing house as the counterparty that does not fail, which is why central clearing improves on bilateral trading. Here the exchange and its CCP faced the prospect of member defaults they judged could exceed the resources standing behind the clearing house, and the exchange's resolution was to void executed trades. No member defaulted, and the default waterfall was never drawn on.

So the risk that materialized was not the price. A trader who was long nickel, correct, and fully margined had the profit taken away, which the reader met in a different market as

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

Section 37.10's auto-deleveraging, and which Chapter 2's account of this episode records as trade finality yielding to the clearing house's survival.

A CCP's guarantee is bounded by its resources and its governance, and when they are threatened it can become a negotiation; Section 2.8's waterfall has a bottom, and this case showed that an exchange may act before its clearing house gets there. A member or end user should measure each layer of that waterfall against a move like March 2022's and read the venue's rules on canceling trades with it, which is Section 2.10's question of who absorbs the loss; the venue should know the full size of a large position, over-the-counter holdings included.

40.8 UK LDI: THE HEDGE THAT CONSUMED ITS OWN COLLATERAL

In September 2022 a rapid rise in gilt yields forced UK pension schemes into a self-reinforcing sale of the assets backing their hedges. What makes the case valuable is that the schemes' hedges did their job.

Example 40.5 Better off and unable to pay

Take a system with roughly £1.5tn of liabilities at a duration near twenty, hedged with leveraged gilt and swap positions requiring variation margin. The schemes' reported hedge ratios were higher than the 40% used here, but part of each hedge was gilts held outright, which call for no collateral. Take the margined part as covering 40% of the liabilities' rate sensitivity, so each margin call is 40% of the fall in the liabilities' value:

Yield rise	Liability falls by (duration held at 20)	Margin call
50 bp	10%	£60bn
100 bp	20%	£120bn
150 bp	30%	£180bn
250 bp	50%	£300bn

The funding position improved at every row. Because the hedge covered only part of the liabilities' rate sensitivity, the liabilities fell faster than the assets, so the schemes became *better funded* as the crisis progressed—while unable to meet the margin calls.

Selling gilts to raise the collateral pushed yields higher, which raised the next margin call. The hedge's own success generated the demand that forced its unwinding.

This is Section 8.9 in its purest form—solvency and liquidity are different quantities, and only one of them forces a sale—and the lesson is Section 40.1's. The margined hedge in Example 40.5 is a DV01 of $0.40 \times £1.5\text{tn} \times 20 \times 0.0001 = £1.2$ billion per basis point, so Example 8.10's arithmetic makes a 100 basis point move a £120 billion margin call, and Section 38.17's reverse stress test, run on the buffer a scheme holds, finds the move that exhausts it—the figure to report. Plan for a larger move than any one scheme's stress, because forced gilt sales raise yields again: Example 8.11's loop, whose total only a supervisor can see.

40.9 AIG: A COLLATERAL TRIGGER ON A POSITION THAT NEVER DEFAULTED

By the end of 2007, according to AIG's own annual report, its financial products unit had sold protection on about $527 billion of super-senior tranches, on a view that they attached above any plausible loss. Most of it was written for European banks seeking regulatory capital relief; the margin calls came mainly from about $78 billion written on multi-sector CDOs backed largely by subprime mortgages. The view was defensible on realized defaults for much of the book, and it was irrelevant, because the contracts contained two features the view did not price:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

mark-to-market collateral against spread widening, and additional collateral on a downgrade of the seller.

Spread widening	Collateral demanded on $78bn (duration held at 5)
100 bp	$3.9bn
200 bp	$7.8bn
400 bp	$15.6bn
800 bp	$31.2bn

No row requires a default. The position was a view on realized losses; the contract was a claim on marks, and Section 25.4's mark moves with the spread, default or not.

The downgrade trigger is unlike anything in an ordinary hedge, whose cash demands depend on the market and not on the hedger's own credit. It converts a deterioration in the *seller's own* condition into an immediate cash demand, so the contract becomes most expensive exactly when the seller is least able to pay. That is Section 27.9's wrong-way risk written into the contract rather than arising from a correlation. In September 2008 the downgrade came, the calls could not be met, and the Federal Reserve lent AIG $85 billion.

Price the collateral terms, not only the view on defaults. Section 25.4's mark makes the collateral a duration calculation—$78 billion at a duration of 5 posts $39 million for every basis point of widening, with no default—so the seller's liquidity should be sized by Section 8.10's stress test applied to spreads, not by expected loss. The downgrade trigger belongs in the same test, assumed to arrive with the widening that produces it.

40.10 THE LONDON WHALE: THE MEASURE CHANGED AND THE RISK DID NOT

In 2012 a credit index book inside a bank's treasury function lost about $6.2 billion. The instructive part happened months earlier and cost nothing at the time.

Reported VaR under the previous model	$132m
Reported VaR under the new model	$66m
Reduction	**50%**
Positions changed on the day	**none**

A model change, hastily approved and containing a spreadsheet error, halved the desk's reported risk without anyone trading. Over the following months the book's notional roughly tripled under the looser limit. The eventual loss was 47 times the old figure and 94 times the new one—a one-day figure set against a cumulative loss, but the scale is the point.

This is Chapter 39's model risk in its purest administrative form. Section 39.14 argued that an error in the model misstates the mark and the measure together; here the measure moved alone, and the limit framework built on it loosened accordingly. A limit expressed in units of a model is a limit the model can relax.

This case also most directly justifies Section 39.6's attribution test. The desk's risk model stopped describing its book, and the mechanism that now exists to detect that (comparing the P&L the risk model produces with the desk's own revaluation, day by day) is exactly the test this case calls for. Whether it would have caught this one is unclear, since Example 39.5 shows a desk can leave about a third of its P&L variance unexplained and still pass.

Validate a VaR model change as Section 39.14 validates any model, before the new number reaches a limit: run old and new side by side, require the fall from $132 million to $66 million on an unchanged book to be explained, and ask for Section 38.16's backtest on both. A limit the model cannot relax, on gross notional, should sit beside the VaR limit, so that a book tripling in size is visible whatever the measure says.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

40.11 COMPLEXITY SOLD TO A BUYER WHO COULD NOT PRICE IT

A fifth mechanism runs through several cases and is different in kind from the four that Section 40.15 draws from the cases so far, because the loss is not caused by cash, size, information, or a venue. It is caused by an asymmetry in who can value the trade.

Example 40.6 Two leverages multiplied

Orange County, California, ran a $7.5 billion municipal investment pool, levered it to about $20.5 billion through repo (**2.73**×) and invested a large share of it in **inverse floaters**, whose coupon falls as rates rise.

The instrument carries its own leverage. An inverse floater with coupon leverage L, paying a fixed rate less L times the floating rate, has roughly $(1 + L)$ times the duration of its underlying, taken here as 4 years, so the two multiply. Were the whole book in such notes:

Coupon leverage	Instrument duration	Portfolio effective duration
1	8.0 yr	21.9 yr
2	12.0 yr	**32.8 yr**
3	16.0 yr	43.7 yr

The actual book mixed inverse floaters with other structured notes and plain agency paper, so its duration was lower than any row above. Its 1994 loss of about $1.64 billion, on a rise of roughly 300 basis points in short and intermediate yields, corresponds to an asset duration of $1.64/(0.03 \times 20.5) \approx 2.67$ years, an effective duration of $1.64/(0.03 \times 7.5) \approx 7.3$ years on the depositors' $7.5 billion:

Rate rise	Loss	Share of the $7.5bn
100 bp	$0.55bn	7.3%
200 bp	$1.09bn	14.6%
300 bp	$1.64bn	**21.9%**

Nothing was concealed. The coupon formula was in the term sheet and the repo was on the balance sheet. The pool reported its yield advantage rather than its duration, warnings during 1994 that further rate rises threatened it were dismissed, and the product was described, accurately as far as its yield went, as an income enhancement. The county filed for bankruptcy in December 1994.

The same asymmetry produced the structured-swap disputes of the mid-1990s, in which corporate treasurers bought leveraged rate positions embedded in what were presented as financing trades, and later argued in court that they had not understood what they held. The legal outcomes varied; the mechanism did not.

It is not a historical category. Chapter 35's retail structured products carry the same asymmetry in a regulated wrapper, Section 35.10's estimated initial value exists precisely to quantify it, and Section 37.8's liquidity providers are selling a straddle described to them as yield. The mechanism persists because the asymmetry is structural: the party that constructs a payoff is usually better placed to price it than the party that buys it.

The response to this category is disclosure of the computed risk, not only its inputs, and suitability, because no risk system helps a buyer who does not know which risk to measure. The measure Orange County needed was Section 4.11's duration taken through the whole structure: about 2.67 years on the assets times repo leverage of 2.73 is Example 40.6's 7.3 years on the depositors' money, so each 100 basis points of rise costs 7.3% of the pool. A buyer should not hold a payoff whose duration it cannot compute, and Sections 35.11 and 35.12 set out how a note is evaluated and what disclosure cannot fix.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

40.12 EUROPEAN ENERGY 2022: THE SAME QUARTER, A DIFFERENT INDUSTRY

The LDI episode was not the only funding crisis of that quarter, and the second one matters because it suggests the mechanism is structural rather than a feature of pension funds. A generator that has sold power forward holds a short position that is margined daily. The plant that will deliver the power is not. Example 8.9 works the arithmetic for one producer; at the scale of a large generator, 50 TWh of forward sales and a move from 50 to 400 EUR/MWh demand EUR 17.5 billion of cash against a position whose entire purpose was to remove risk. The generator's value rose throughout (its plant's unsold output now sells into a far higher market), yet several such firms required state liquidity support to avoid failing. The support went to the firms that had hedged: those that had not faced no margin calls and took windfall profits as prices rose, which is what a system produces when it prices risk correctly and liquidity not at all.

This case bears on a question the earlier cases leave open. Metallgesellschaft could be dismissed as a strategy error and the LDI schemes as a leverage error. Two simultaneous instances in unrelated markets suggest the exposure is created by the market structure: it arises wherever a hedge is margined and the hedged item is not. Neither 2022 case was a product of mandatory clearing: most of the utilities hedged on exchanges by choice, and the pension schemes, exempt from the clearing mandate, met margin calls on repo and bilaterally margined swaps. But the post-2008 push toward margining, voluntary and mandatory, that Chapters 2 and 8 describe widened the footprint of that shape.

The lesson is Section 40.1's again, at the level of an industry. A forward sale of 50 TWh posts EUR 50 million of variation margin for every euro the power price rises, so the liquidity a generator commits before hedging should be sized by Section 8.10's stress test on a move like 2022's, not a typical month. A policymaker who pushes hedging into margined form should ask where that liquidity comes from in a stress.

40.13 MARK-TO-MODEL, AND POSITIONS WITH NO MARKET

A further mechanism applies to positions with no market price. Section 39.11's prudent valuation exists because the value carried for them is produced by the holder; this section adds the cases.

The failure mode is not fraud, though fraud is available. It is that a valuation that cannot be checked is a valuation that responds to pressure, and the pressure is asymmetric: no one objects to a mark that produces a profit.

The structural cases are two: Enron's long-dated energy contracts, booked at their full modeled value on the day of signing, which converted a decade of anticipated profit into current-period earnings; and illiquid structured credit carried at model prices while the market for it disappeared. In both, the position and the price came from the same place.

The safeguards are Chapter 39's. Independent price verification exists because the desk cannot be the source of its own marks. The model risk component of Example 39.9's adjustment, the largest single item, is a deduction for precisely this, and it is taken whether or not anyone suspects a problem.

The connection to machine learning is direct. A neural-network pricer's silent extrapolation outside the range of its training data (the training box of [TN41]) is mark-to-model with the model's confidence removed from view, which is why Section 39.15 insists the training set be recorded as part of the model.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

40.14 SILICON VALLEY BANK: THE LOSS FROM NOT HEDGING

This is the most important case in the chapter. A bank funded by concentrated, uninsured, rate-sensitive deposits held a large securities portfolio at meaningful duration and left the interest rate risk unhedged, having unwound during 2022 most of the pay-fixed swaps it had held against part of the book.

Example 40.7 What a swap would have cost, and what its absence did

Take a $120bn portfolio at duration 5.6:

Yield rise	Economic loss (duration held at 5.6)	Share of a $16bn equity base
100 bp	$6.7bn	42%
200 bp	$13.4bn	84%
300 bp	$20.2bn	**126%**
400 bp	$26.9bn	**168%**

A pay-fixed swap on about half the duration would have cost negative carry (in 2021, paying a fixed rate near 1% against an overnight rate near zero, on the order of 100 basis points a year on the swapped notional) and would have removed about half of that column. The instrument was standard, liquid, cleared, and priced by Chapter 7's arithmetic. Nothing about the hedge was difficult.

Accounting completed the failure. Securities classified as held-to-maturity are not marked through capital, so the reported number stayed intact while the economic number did not. The unrealized loss, about $15 billion at the end of 2022, was visible in the footnotes but not in capital, until, in March 2023, a loss-making sale of almost all its available-for-sale securities and a capital raise put it in front of depositors at once, by showing that the bank needed cash.

So the sequence is: an unhedged exposure, an accounting treatment that concealed it, and a funding base that could leave in a day. Each is necessary; none alone is sufficient.

A bank failed because it did not hedge its rate risk, and the derivative it did without was the plainest instrument in this book. Section 3.2's case for hedging applies in full, and it is worth teaching against the reflex that derivatives create danger.

Banking-book rate risk sits outside Chapter 39's trading-book rule. The Basel Committee's 2016 standard treats it in Pillar 2, with a supervisory outlier test: a change in the economic value of equity, under prescribed rate shocks, above 15% of Tier 1 capital. It looks through the reported number to the economics, but the US agencies never adopted it as a rule, and it did not bind SVB.

Hedge the economic duration with the tools of Sections 4.11 and 6.7: $120 billion at a duration of 5.6 is a DV01 of $\$120\text{bn} \times 5.6 \times 0.0001 \approx \$67$ million against $16 billion of equity, and a pay-fixed swap valued by Section 7.3's arithmetic and sized to half of it would have halved every row of Example 40.7's table. Measure that DV01 with held-to-maturity securities included, whatever the accounting says: as Section 3.8 shows, the swap would have been marked through earnings while the held-to-maturity securities were not, so the reported numbers would have penalized the hedge.

40.15 WHAT REPEATS

Across the cases of this chapter, four mechanisms recur and the instruments do not.

Cash and value are different quantities. Metallgesellschaft, the LDI schemes, the 2022 European generators, AIG, and Example 37.2's basis trade are one mechanism: a position correct in value can be closed for want of cash, and no valuation model contains the closing.

Size and leverage change the exit. LTCM, Amaranth, Archegos, and the nickel case all failed at the point of liquidation rather than at the point of valuation. The measure describes the

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

position's distribution; the loss came from the position's share of the market, or from leverage that made the path decisive, which are different variables.

Someone must see the whole. Barings, Société Générale, and Archegos are cases where the aggregate existed and no one charged with controlling it saw it (inside a firm in the first two, across firms in the third). The failure is informational, and in Archegos the instrument enabled it.

And the guarantee has a bottom. The nickel case and Section 37.10's auto-deleveraging both show that the mechanism meant to make a market safe is bounded: at the exchange, trades were voided before any resource was drawn on, and at a crypto venue, profitable positions are closed once the insurance fund is gone.

A fifth mechanism sits apart. Section 40.11's asymmetry is not a failure of measurement but of *who is equipped to measure*, and it is the only one here that recurs in regulated retail form. Section 40.13 adds a sixth: a mark that only the holder can produce.

Case Study 40.1 The information was already inside the firm

The most consistent finding across the official inquiries is also the least comfortable, because it rules out the easiest explanation, that no one could have known.

In almost every case the facts needed to prevent the loss existed inside the institution before it happened. Barings' reconciliation discrepancies were reported and unresolved for months. The exchange-traded part of the nickel short was visible to the exchange, though much of the position was held over the counter. The credit index desk's own valuations were disputed internally before the loss was recognized. The unhedged securities portfolio of Section 40.14 had been the subject of supervisory findings. The Archegos post-mortem commissioned by Credit Suisse is explicit: the bank's risk function had escalated the concentration, and the escalation did not change the position.

So the failure is rarely detection and usually escalation. Someone saw it; the observation did not reach a person with both the authority and the incentive to act, or reached them in a form that competed with a revenue argument and lost.

That is why so little of the reform in Section 40.16 is quantitative. Segregating trading from settlement, requiring mandatory leave, making risk report outside the business line, and giving a desk-level test a capital consequence are all attempts to change what happens to an observation, not to generate a better one.

The corollary for a reader of this book is uncomfortable. Improving a measure has no effect if the existing one was already overridden, and much of Chapter 39's machinery, the backtesting zones and the attribution test with their capital consequences, tries to make overriding expensive rather than measurement better. That is the right design.

40.16 WHAT ACTUALLY CHANGED

The reforms these cases produced follow an unflattering pattern. The control failures produced durable change. Segregation of trading from settlement, independent reconciliation, and mandatory leave became standard after Barings, and such losses are rarer, though Société Générale in 2008 and UBS in 2011 (a $2.3 billion unauthorized-trading loss) showed that controls are not self-enforcing. This, the profession's clearest success, required no modeling.

The clearing reforms after 2008 were substantial and are examined in Chapters 2 and 27. They belong to a wider shift, though, which also widened the exposure of Sections 40.8 and 40.12: margining more of the market, voluntarily as well as by mandate, converts credit risk into liquidity risk, and the 2022 LDI and energy episodes showed the cost of that conversion.

The measurement failures produced the least. Amaranth's lesson about concentration was

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

available in 2006, and Archegos repeated it in 2021 with a different instrument. Chapter 39's framework prices illiquidity and still does not measure concentration.

The accounting problem behind Section 40.14 is unresolved. The treatment that concealed the loss remains available, and the arguments for it are not frivolous: marking a held-to-maturity book through capital would import volatility into the accounts of institutions that genuinely intend to hold. The gap is real and the fix is not obvious, so the case will recur.

40.17 WHAT INSTITUTIONS AND REGULATORS SHOULD TAKE AWAY

Above the cases sits one rule: separate the instrument from the failure. In every case but nickel, where the venue voided the contract, the derivative did what its contract said, and a risk function that blames the instrument will look for the next problem in the wrong place. The failures were in cash, size, information, valuation, and governance, and the advice follows them.

Six points apply to a financial institution:

- *Measure liquidity beside value.* Report, for every hedged position, the margin it can call under stress against the liquidity committed to meet it, gross and never netted against the hedged item (Metallgesellschaft, LDI, AIG, the 2022 generators).
- *Size positions by their exit.* Limit the share of open interest and the client's whole position across brokers, not only the notional or the VaR, and stress the exit with a price-impact cost (Amaranth, Archegos, nickel).
- *Cap what a control failure can cost.* Keep the segregation of duties, and cap the margin funding any one desk can draw, so that the largest position that could be hidden is small against equity (Barings, Société Générale).
- *Separate who marks from who profits.* Verify prices independently, reserve for model risk, and validate a risk-model change before it sets a limit (the London Whale, Section 40.13).
- *Start from the failure.* Run a reverse stress test for the move that exhausts equity or liquidity, and assume others hold the same position (LTCM, portfolio insurance, LDI).
- *Treat removing a hedge as a decision.* Leaving a risk unhedged is a position, and taking a hedge off deserves the review that putting it on received (SVB).

Six apply to a regulator:

- *Look at the aggregate.* Margin can be adequate in every bilateral relationship and inadequate in total; position reporting across prime brokers and supervisory review of concentration cover what no single firm sees (Archegos, Amaranth).
- *Supervise liquidity as closely as capital.* Margining converts credit risk into liquidity risk, so buffers should be sized to stressed margin calls, including at non-bank users such as pension schemes and utilities (LDI, the 2022 generators).
- *Interrupt feedback rather than ban strategies.* Where many holders follow the same rule, circuit breakers, margin that does not fall in calm markets, and backstops prepared in advance cut the loop (portfolio insurance, LDI).
- *Make a venue's guarantee, and its limits, known.* Test default resources against plausible extreme moves, and publish the rules for canceling trades before they are needed (nickel).
- *Protect the buyer who cannot price.* Require disclosure of duration, leverage, and a payoff scenario table, and a suitability test on the payoff's shape (Orange County).
- *Look through the accounting.* Supervisory measures should capture economic duration and unrealized losses, whatever the accounting classification (SVB).

Five apply to governance and culture, at institutions and supervisors alike:

- *Question an unexplained profit as hard as an unexplained loss.* A book reported as

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

low-risk arbitrage that earns like a directional one is directional, and no one objects to a mark that produces a profit (Barings, Société Générale, Section 40.13).

- *Fix escalation, not only detection.* Give the risk function an independent line to the chief executive and the board, and do not let the desk settle a risk manager's objection, because the concern had usually been raised and lost (Archegos, the London Whale).
- *Pay as the risk is realized.* Defer and claw back pay, and judge a desk on profit net of the risk and capital it uses, so that neither a decade of modeled profit booked at signing nor a halved VaR is rewarded (Enron, the London Whale).
- *Approve a position again as it grows.* A book approved at one size is a different book at three times that size, so growth past a set threshold should send it back to whoever approved it (the London Whale, Archegos).
- *Name an owner for each risk, and protect whoever raises it.* A discrepancy reported and left unresolved for months needs a named senior manager answerable for closing it, and the person who reported it needs protection from the revenue argument it contradicts (Barings, the case study in Section 40.15).

SUMMARY

In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed; what failed was around them: the cash, the exit, the information, the venue, the buyer's ability to price, or the decision not to hedge at all. Across the cases four mechanisms recur, and the instruments do not.

Cash differs from value. Metallgesellschaft's stack hedge was flat in value at every price and catastrophic in cash, because margin is paid now and the supply profit arrives over years. The UK LDI schemes and the European generators met the same shape in the same quarter of 2022, a hedge margined daily against a hedged item that is not, which suggests the exposure is created by market structure. AIG met it in another form, with collateral called on spread widening and no defaults at all.

Size and leverage change the exit, and someone must see the whole. LTCM, Amaranth, and Archegos failed at the point of liquidation rather than of valuation. Archegos's prime brokers each margined their own slice of a position no one saw whole; at Barings and Société Générale a trader defeated the controls, and the ratio of the hidden position to equity decided which firm survived. Portfolio insurance in 1987 was the market-wide version: a hedge rule shared widely enough becomes the flow and deepens the decline it insures against.

The guarantee is bounded, and measures can bend. When nickel more than tripled in two days the exchange voided executed trades, so a central counterparty's guarantee depends on its resources and its governance. An approved model change halved the London Whale's VaR with no position changed. A fifth mechanism sits apart: Orange County could not price what it bought, though nothing was concealed.

The most recent case reverses the usual story. Silicon Valley Bank failed because it left its rate risk unhedged, so a reading that treats derivatives as the source of danger has the sign wrong. The lesson is to separate the instrument from the failure. In most cases the facts needed to prevent the loss already existed inside the firm, and what failed was escalation. The control failures produced durable reform; the measurement failures produced the least.

FURTHER READING

Most of these events produced an official inquiry or a commissioned review, and those reports are the primary sources. On Barings, the place to start is the Board of Banking Supervision's 1995 inquiry; on 1987, the Brady Commission's report of 1988, which Section 16.2 also draws on. On the London Whale, JPMorgan's own Management Task Force report and the

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

US Senate Permanent Subcommittee on Investigations' case history, both of 2013, are worth reading together, since one is the bank's account and the other a legislative investigation of it. On Archegos, the Paul, Weiss report prepared for Credit Suisse (2021) is the primary source, particularly on what the bank's own risk function reported while the position was building. For the 2022 episodes, Oliver Wyman's independent review of the nickel suspension, prepared for the LME (2023), and the Bank of England's *Financial Stability Report* of December 2022 on the LDI intervention both state what the authorities did not know at the time, which is the part worth reading. On Silicon Valley Bank, the Federal Reserve's April 2023 review of its own supervision is the source, and it is explicit about examination failures. On AIG, the Financial Crisis Inquiry Commission's report (2011) sets out the collateral calls and the downgrade. On Metallgesellschaft, Culp and Miller (1995) and Edwards and Canter (1995) are worth reading for the disagreement rather than the conclusion: whether the hedge was sound depends entirely on whether one counts the funding requirement as part of the strategy, and the two sides of that argument are the two halves of Section 40.1.

PRACTICE QUESTIONS

40.1 Using Example 40.1, explain why the stack hedge is flat in value and catastrophic in cash, and identify which leg produces the cash demand.

40.2 Using Figure 40.1, state the peak funding gap and when it occurs, and explain why the strategy can fail between the two points even though both curves end at the same total.

40.3 Explain how a stack-and-roll converts a flat-price hedge into a calendar spread, and state the first-year cost in Example 40.1's contango case.

40.4 Explain what Barings and Société Générale have in common, and why the same failure destroyed one firm and not the other.

40.5 Using Example 40.3, compute the move that exhausts LTCM's equity, and explain why this makes the strategy a view about the path rather than about value.

40.6 Using the Amaranth figures, compute the days required to exit at a 10% participation rate, and explain why the measure's two assumptions, its one-day horizon and prices unmoved by the seller, fail simultaneously.

40.7 Using Example 40.4, explain why each prime broker could judge its own slice adequately margined while the system was not.

40.8 Explain what assumption of Chapter 2 the nickel case tested, and what the risk that materialized actually was.

40.9 Using Example 40.5, explain how a scheme's funding position could improve at every step while it was unable to meet collateral calls.

40.10 Explain why AIG's collateral demands required no defaults at all, and what distinction of Chapter 25 the case turns on.

40.11 Explain what changed and what did not change when the credit index desk adopted a new VaR model, and state the loss as a multiple of each reported figure.

40.12 Using Example 40.6, explain how portfolio leverage and instrument leverage multiply, infer the pool's effective duration from its $1.64 billion loss on a 300 bp rise, and compute the loss on a 200 bp rise at that duration.

40.13 Explain why nothing was concealed in the Orange County case and the buyer still lost, and what this implies about the limits of disclosure.

40.14 Explain why a European generator that had sold power forward faced margin calls while

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

its economics improved, and compute the demand on 50 TWh for a move from 50 to 400 EUR/MWh.

40.15 Explain why a valuation that cannot be independently checked responds to pressure, and why the pressure is asymmetric.

40.16 Explain the case study's finding that the failure is rarely detection, and why so little of the resulting reform is quantitative.

40.17 Using Example 40.7, compute the economic loss on a $120bn portfolio at duration 5.6 for a 300 bp rise, and state it as a share of a $16bn equity base.

40.18 Explain the role of held-to-maturity accounting in the same case, and why a loss disclosed in the footnotes never reached capital, yet a sale and a capital raise put it in front of depositors.

40.19 Name the four mechanisms that recur across these cases, and give one case for each.

40.20 Explain which category of failure produced durable reform and which produced the least, with an example of each.

40.21 Using Example 40.2 and Figure 40.2, compute the futures the programs must sell as the index falls from 100 to 90, and explain why the selling per index point peaks near the floor and slows once the delta approaches -1.

FURTHER QUESTIONS

40.22 Example 40.1 separates the value and cash profiles of a hedge. Construct a general measure of a hedge's funding requirement (the distribution of cumulative cash demanded before the offsetting gain is realized) and apply it to the LDI position of Example 40.5.

40.23 Example 40.4 shows margin adequate bilaterally and inadequate in aggregate. Design a reporting mechanism that would have revealed the total, state what each party would have to disclose and to whom, and determine whether the disclosure would itself have moved the market.

40.24 Section 40.5 treats exit time as a function of position share. Build a liquidation-cost model with price impact proportional to participation, determine the optimal unwind schedule for a position at 40% of open interest, and compare the resulting cost with the position's reported VaR.

40.25 The nickel case ended in trade cancellation. Design a CCP recovery arrangement that avoids voiding executed trades, state who funds it, and determine whether any participant would rationally join a clearing house on those terms.

40.26 Section 40.14 argues the accounting treatment concealed the exposure. Model a bank's optimal hedging decision under held-to-maturity accounting versus full fair value, and determine the conditions under which the accounting choice alone flips the hedging decision.

40.27 Section 40.16 argues that measurement failures produced the least reform. Take the concentration gap specifically, propose a supervisory measure that would capture it, and determine what data would be required and from whom.

40.28 Every case in this chapter was reconstructed after the fact. Choose one and identify the earliest point at which the information available inside the firm would have been sufficient to act, then state what report would have had to exist for someone to have seen it.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!